

Modeling of Water Transfer Delay for Optimal Operation of Integrated Electrical-Water System

Yiming Yao , Chunyan Li , *Member, IEEE*, Heng-Ming Tai , *Senior Member, IEEE*,
and Changzheng Shao , *Member, IEEE*

Abstract—This paper investigates the transfer delays of a water distribution system (WDS). The delay implies that water balance between the supply and demand does not meet instantly, which may affect the regulation capability of WDS on the electricity consumption profile. Current research did not consider the role the transfer delay plays in the cooperative operation of power system and WDS. Not considering the delay hinders the water flow dynamics from being used to provide energy flexibility for the power system operation. An electrical-water analogy method is proposed to model the water transfer delay of WDS. The cooperative operation model of the WDS and power system is created to facilitate the incorporation of wind power to the power system and lower the operation cost. Moreover, a graph-based searching scheme is developed to enable efficient and accurate calculation of the pressures and water flows in the WDS network. Case studies are presented to demonstrate the merits of the proposed electrical-water analogy method in power system operation.

Index Terms—Power system, water distribution system, transfer delay, electrical-water analogy, wind power.

NOMENCLATURE

A. Sets

$\Omega^T / \Omega^W / \Omega^{WDS}$	Set of conventional generating units/wind farms/WDSs.
Ω^L	Set of blocks in the piecewise function of the conventional generating unit.
Λ^E / Ψ^E	Set of electrical buses/lines.
$\Omega_n^T / \Omega_n^W / \Omega_n^{WDS}$	Set of units/wind farms/WDSs that connected to bus n .
Λ_n^E	Set of electrical buses connected to bus n .
$\Delta_P / \varpi / \Delta$	Set of fixed-speed pumps/water nodes/pipes.

$\mathcal{F} / \mathcal{L}$

$\mathcal{O} / \mathcal{R}$

$\mathcal{H}$

B. Parameters

g

D/A

α

f

$\text{sign}(Q)$

Q_b

$A_1^i(s), A_2^i(s)$

$B_1^i(s), B_2^i(s)$

H_{min}

Q_{min}/Q_{max}

L_i

t_s/t_e

$q_{P_m}^{min}/q_{P_m}^{max}$

$a_{P_m}, b_{P_m}, c_{P_m}$

$a_{S_m}, b_{S_m}, c_{S_m}$

V_b

$\hat{\omega}_{S_m}$

$\omega_{S_m}^t$

$q_{P_m,b}^t/q_{S_m,b}^t/q_{V_m,b}^t$

$\kappa_1, \kappa_2, \kappa_3$

N_h^{WP}

k_V

Set of reservoirs/demand nodes in the WDS.

Set of directed paths that go through cross nodes/do not go through cross nodes.

Set of cross nodes.

Acceleration of gravity.

Diameter/cross-sectional area of the water pipe.

Velocity of the water hammer wave.

Friction factor.

Sign function which denotes the direction of flow in the pipe. It equals to 1 if $Q > 0$, zero when $Q = 0$, and -1 when $Q < 0$.

Baseline flow of the pipe.

Algebraic expressions in the Laplace domain.

Algebraic expressions in the Laplace domain.

Required minimum head pressure along the pipe.

Minimum/maximum volume of the pipe.

Length of pipe i .

Starting time/ending time.

Minimum/maximum value of the flow when fixed-speed pump m on.

Coefficients in the head-flow equation of fixed-speed pump m .

Coefficients in the head-flow equation of variable-speed pump m .

Baseline value of the variable in the hydraulic model of WDS components.

Maximum allowable relative speed of the variable-speed pump m .

Baseline speed of the variable-speed pump m at time t .

Baseline flow of fixed-speed pump/variable-speed pump/valve m at time t .

Weighting coefficients.

Number of pumps in WDS h .

Coefficient in the head-flow equation of the valve.

Manuscript received 2 July 2021; revised 2 November 2021 and 7 February 2022; accepted 13 March 2022. Date of publication 16 March 2022; date of current version 22 December 2022. This work was supported by the National Natural Science Foundation of China under Grant 52177073. Paper no. TPWRS-01059-2021. (*Corresponding author: Chunyan Li.*)

Yiming Yao, Chunyan Li, and Changzheng Shao are with the State Key Laboratory of Power Transmission Equipment and System Security, Chongqing University, Chongqing 400030, P.R. China (e-mail: yao_cqu@163.com; lcy-cqu@cqu.edu.cn; cshao@cqu.edu.cn).

Heng-Ming Tai is with the Department of Electrical and Computer Engineering, University of Tulsa, Tulsa, OK 74104 USA (e-mail: tai@utulsa.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPWRS.2022.3159786>.

Digital Object Identifier 10.1109/TPWRS.2022.3159786

$q_{V_m}^{min}/q_{V_m}^{max}$	Minimum/maximum value of the flow when valve m on.
Δt	Time step.
$A_{J_m}^{tank}/\mu_W$	Surface area of tank m /water loss rate of the tank.
$W_{J_m}^{min}/W_{J_m}^{max}$	Minimum/maximum water storage of tank m .
w_K/w_h	Water demand at node K/h .
w_C	Water demand at the cross node.
N/M	Number of nodes/branches in the WDS.
$a_{h,k}$	Connection relationship between the node n_h and the branch l_k .
$c_i/c_i^u/c_i^d$	Commitment/startup/shutdown cost of unit i .
N_g/T	Number of conventional units/time steps.
N_W/N_{WDS}	Number of wind farms/WDSs.
λ	Penalty factor for wind power curtailment.
L	Number of piecewise linear cost function segments.
$K_{i,l}$	Slope of block l of the piecewise function of unit i .
TU_i/TD_i	Minimum up/down times of unit i .
$Ramp_i/\lambda price_t$	Ramp rate of unit i /electricity price at time t .
P_i^{gmin}/P_i^{gmax}	Minimum/maximum output of unit i .
$P_{w,t}^{pre}$	Output of wind farm w at time t .
$d_{n,t}$	Demand of electrical load n at time t .
$B_{n,m}$	Susceptance of electrical line $n-m$.
$PF_{n,m}$	Capacity limit of electrical transmission line $n-m$.

C. Variables

H	Head pressure.
v/Q	Velocity/volume of flow in the pipe.
ΔQ	Difference between the optimized and the baseline values of the flow in the pipe.
$\hat{h}_{P_m}^t/\hat{h}_{S_m}^t/\hat{h}_{V_m}^t$	Pressure head gain of fixed-speed pump/variable-speed pump/valve m at time t .
$q_{P_m}^t/q_{S_m}^t/q_{V_m}^t$	Flow in fixed-speed pump/variable-speed pump /valve m at time t .
$\omega_{S_m}^t$	Ratio of actual speed to the nominal speed of the variable-speed pump m at time t .
$P_{i,t}^{WP}$	Electricity consumption of pump i at time t .
$P_{h,t}^{WDS}$	Electricity consumption of WDS h at time t .
$q_{J_m,t}^{in}/q_{J_m,t}^{out}$	Water that flows into/out of tank m at time t .
$W_{J_m,t}$	Volume of the stored water in tank m at time t .
$h_{J_m}^t$	Head pressure of tank m at time t .

V	Generic variable used to represent the variables of WDS components. For the water pipe, V refers to Q . For the fixed-speed pump, V refers to $q_{P_m}^t$. For the variable-speed pump, V refers to $q_{S_m}^t$ and $\omega_{S_m}^t$. For the valve, V refers to $q_{V_m}^t$.
ΔV	Difference between the optimized and the baseline values of the variable in the hydraulic model of WDS components.
$\Delta q_{S_m}^t/\Delta \omega_{S_m}^t$	Difference between the optimized and the baseline values of the water flow/relative speed in the variable-speed pump m at time t .
Q'_K	Calculated outlet flow of the pipe whose end node is node K .
$Q_{in,C}/Q_{out,C}$	Total inlet/outlet flow of the cross node.
$H_{C,n}$	Inlet/outlet pressure of pipe n whose inlet/outlet node is the cross node.
$P_{i,t,l}^g$	Power produced in block l of the piecewise linear function of the unit i at time t .
$\theta_{n,t}/\theta_{ref,t}$	Phase angle of electrical node n /reference node at time t .
$I_{i,t}/U_{i,t}/D_{i,t}$	Binary variable for commitment/startup/shutdown status of unit i at time t (equals to 1 when the unit is on/started up/shut down, and 0 otherwise).
$P_{w,t}^{wc}$	Amount of wind power curtailment in the wind farm w at time t .
Q_{nh}/Q_{hm}	Flow in pipe $n-h/h-m$.

I. INTRODUCTION

Water distribution system (WDS) accounts for 5% of the annual total electricity consumption in China, and up to 7.1% in California, USA [1], [2]. Electrical demand associated with WDS is projected to increase. WDS can be regarded as an adjustable electrical load from the perspective of power system [3]. The electricity consumption profile of WDS can be regulated flexibly by coordinating the operation of water pumps and tanks. The cooperative operation of the power system and WDS attracts great attention recently due to its merits of high-efficiency, cost-effectiveness and sustainability [4], [5].

The optimal water-power flow problem has been studied to optimize the operation strategies of the power system and WDS considering the network constraints [6]–[8]. Many efforts have been made to deal with the non-convex hydraulic equations of WDS, such as the piecewise linear approximation [9], the second-order cone relaxation [10], and the quasi-convex hull relaxation method [11]. The energy flexibility of WDS has been analyzed and integrated into the operational model of power system [12], [13]. The controllable energy assets such as the water pump equip the WDS to provide the demand response service and the frequency regulation service in the electricity market. The models for the WDS to bid in the electricity market were also developed to maximize the profit [14]–[16]. Most

works on the cooperative operation of power system and WDS assume that the water is transported from the source to the user instantaneously without considering the transfer delays of the water [3]–[16].

To the best of our knowledge, not many researches on the modeling of the transfer delays in the WDS have been reported. Besides, the transfer delays have not been incorporated into the cooperative operation model of the power system and WDS.

The transmission time of the water from the source to the user depends on the velocity of flow and the distance. It can take several hours to transport the water in a large WDS network. Neglecting the transfer delays may have significant impact on the optimal cooperative operation strategies of the power system and the WDS. On the other hand, the transfer delays in WDS offer substantial potential for power system to leverage the flexibility provided by WDS. Electricity travels at a speed close to the light, and it can be assumed that electricity is transported from the generator to the load instantaneously. Thus, the power system must ensure the real-time balance of supply and demand. WDS does not need to instantly balance the supply and demand. The balance can be maintained by the excess water transported from the source to the user. This enables the power shift capability of WDS to be fully leveraged and allows the power system to be operated more flexibly. The shift here refers to the situation that more electricity is consumed during the period of high wind power output and less electricity is consumed when the wind power output is low. More renewable energy can thus be incorporated into the power system and the operation cost of the integrated electrical-water system can be reduced.

This paper proposes a cooperative operation model of the power system and WDS considering the delays of water transmission. The movement of water flow in the pipe can be described by a series of partial differential equations (PDEs) [17]. The finite difference (FD) method, which discretizes the time and space into many points, is used to deal with these PDEs [18]. As the number of discretized points increases, accuracy of the FD method improves, whereas the computational expense increases dramatically. The increasing penetration of renewable energy, such as wind power, has brought great challenges to the power system operation due to its uncertainty. Since the output of wind power varies rapidly, the time resolution of the cooperative operation model should be in minutes. This makes the time-consuming FD method not applicable.

The idea of electrical-water flow analogy has been applied to the modeling of the WDS [19]–[21]. The analogy approach uses an equivalent electric circuit to provide an exact representation of the model. These WDS network models were described by a distributed parameter circuit, which was in the form of PDEs and has to be solved by the FD method. Recently, the method of electric circuit analogy was also exploited in the modeling of integrated energy systems [22]–[26]. The methods in [22], [23] did not consider the transfer delays. The equivalent electric circuit analogy model proposed for a single pipe did not model the transmission process of the flow in the network [24]. Although the methods proposed in [25], [26] were discrete in the time domain, severe calculation errors may occur when the transfer delay is not an integer multiple of the time resolution.

Since the water flow and pressure in WDS vary continuously over the time and space domains, the discrete method may not be applicable.

In this paper, a graph-based electrical-water analogy method is proposed to solve the PDEs of WDS. The method aims to significantly reduce the computation time in comparison to the FD method. The equivalent electric circuit for the WDS is developed and transformed into algebraic equations in the Laplace domain. The water transmission process in WDS is modeled by the graph-based searching method. The inverse Laplace transform can be applied to obtain the flows and pressures along the network in the time domain. The constraints of the water flow and pressure in the pipe expressed in the continuous time and space domains are replaced by a series of simplified constraints.

The main contributions of this paper are

- a) Propose an energy-water model considering water transfer delay in WDS. The proposed model provides a realistic representation of the water flow dynamics which fully exploits the energy flexibility of WDS.
- b) Establish an equivalent electric circuit model to describe the water transfer delay of WDS and provide a series of simplified constraints for the pipe in the continuous time and the space domains.
- c) Develop a graph-based electrical-water analogy method to model the water transmission process in the whole WDS network and an efficient solution methodology to find the solutions efficiently.

In the remainder of this paper, the proposed electrical-water analogy model of the power-WDS system is described in Section II. Rigorous mathematical derivation is presented for verification. Section III presents the detailed modeling of the water transmission process in WDS and the solution procedure for the graph-based method. Section IV describes the cooperative operation model of the power system and WDS. Case studies are given in Section V to demonstrate the benefits of the proposed model on wind power accommodation and computation time reduction. The paper is concluded in Section VI.

II. ELECTRICAL-WATER ANALOGY MODEL OF WATER TRANSFER DELAYS

This section presents the proposed electric circuit analogy model for the water transfer delay in the WDS. The PDEs of the water flow in the pipe are transformed into algebraic equations in the Laplace domain. The non-convex models of the water pump and valve are also turned into linear models using the electrical-water analogy method. Prior to the detailed description of the proposed model, the structures of the WDS and the integrated electrical-water system are outlined.

A. Overview of Electrical-Water System

Fig. 1 depicts the structure of an integrated electrical-water system. The reservoir serves as the source of water for WDS. The water flow is transported from one node to another in the pipe due to the head pressure difference between two nodes. The water pump in the WDS consumes electricity to increase the pressure difference and transport the water flow. More electricity

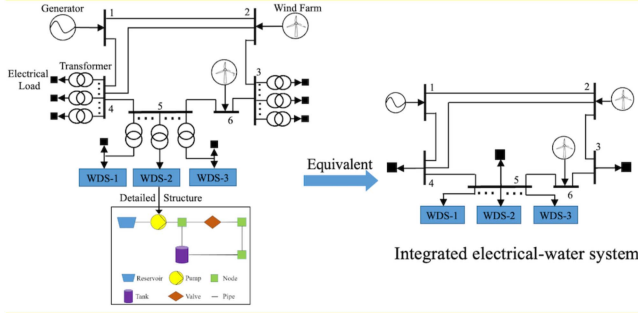


Fig. 1. Example of the integrated electrical-water system.

consumption means that more water can be conveyed. Valves control the pressure in a pipe or the amount of water flowing through a pipe. The water tank can store water or supply water into the system. Control of the water pump and tank equips the WDS to deal with the unbalance of supply and demand in the power system.

This study considers the WDS connected to the power transmission systems for the following reasons.

- 1) The energy consumption of WDS is usually supplied by the power distribution system connected with the load bus of power transmission systems through power transformers. Many researches on the cooperative operation of power system and WDS assume that WDS can be regarded as an equivalent electrical load connected with the load bus of power transmission systems [12], [18], [27]–[29]. As can be seen from Fig. 1, a 6-bus power transmission system supplies electricity to three WDSs linked to bus 5. Then the impact of WDS optimization on the entire power transmission system can be evaluated.
- 2) The large-scale renewable energy output is generally transported and processed via the power transmission system [30]. Since this study considers the WDS optimization on promoting the integration of renewable energy into the power system, the power transmission system would provide better assessment information.
- 3) By replacing the power flow model with a power distribution network, the proposed transfer delay model for WDS also can be applied to the cooperative operation of the power distribution system and WDS. Future research will conduct on a hierarchical control strategy of WDS and the power transmission and distribution systems.

B. Modeling of Water Pipe

The dynamics of water flow are modeled by the motion equation (1) and the continuity equation (2)

$$g \frac{\partial H}{\partial x} + \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{f v |v|}{2D} = 0 \quad (1)$$

$$v \frac{\partial H}{\partial x} + \frac{\partial H}{\partial t} - v \sin \alpha + \frac{\alpha^2}{g} \frac{\partial v}{\partial x} = 0 \quad (2)$$

where v denotes the flow velocity and H the head pressure. They are functions of time t and position x . The above PDEs can be

TABLE I
ANALOGY BETWEEN THE ELECTRIC CIRCUIT AND THE WATER PIPE

Electric Circuit	Water Pipe
Current	Q
Voltage	H
Resistance	$R_b = \text{sign}(Q) f Q_b / (g D A^2)$
Inductance	$L_b = 1 / g A$
Capacitance	$C_b = g A / \alpha^2$
Voltage source	$U_{sb} = \text{sign}(Q) f Q_b^2 / (2 g D A^2)$

further simplified as [17]

$$g A \frac{\partial H}{\partial x} + \frac{\partial Q}{\partial t} + \frac{f Q |Q|}{2 D A} = 0 \quad (3)$$

$$A \frac{\partial H}{\partial t} + \frac{\alpha^2}{g} \frac{\partial Q}{\partial x} = 0 \quad (4)$$

Q is the volume of water flow in the pipe. Equation (3) also holds for the steady flow. Setting $\frac{\partial Q}{\partial t} = 0$ yields

$$\Delta H = - \frac{f Q |Q| \Delta x}{2 g D A^2} \quad (5)$$

which is the widely used Darcy-Weisbach equation in the optimization model of WDS [3]–[16].

The water transfer delay occurs because the flow varies continuously over the time and space as seen in (3) and (4). Given the amount of input flow at the beginning of a pipe, it may take a long time for the output flow at the end of the pipe to be the same as that of the input flow. Equation (5) assumes that the water flow is the same over the pipe in a given time period. Thus, the water transfer delays are not considered.

To solve the PDEs of (3) and (4) in an efficient way, the following electrical-water analogy method is used. For an infinitesimal deviation dx , the corresponding changes in H and Q can be derived from (3) and (4) as given below

$$-dH = \frac{1}{g A} \frac{dQ}{dt} dx + \frac{\text{sign}(Q) f Q^2}{2 g D A^2} dx \quad (6)$$

$$-dQ = \frac{g A}{\alpha^2} \frac{dH}{dt} dx \quad (7)$$

The sign function $\text{sign}(Q)$ is equal to 1 if $Q > 0$, zero when $Q = 0$, and -1 when $Q < 0$. Let $Q = Q_b + \Delta Q$, where Q_b is the baseline flow of the pipe. Then, $Q^2 = (Q_b + \Delta Q)^2 \approx Q_b^2 + 2 Q_b \Delta Q$ or

$$Q^2 \approx 2 Q_b Q - Q_b^2 \quad (8)$$

Substituting (8) into (6) yields

$$-dH = \frac{1}{g A} \frac{dQ}{dt} dx + \frac{\text{sign}(Q) f Q_b Q}{g D A^2} dx - \frac{\text{sign}(Q) f Q_b^2}{2 g D A^2} dx \quad (9)$$

Table I presents an analogy between the electric circuit and the water pipe. The equivalent electric circuit for the pipe is depicted in Fig. 2.

Applying Laplace transform to (7) and (9) with respect to time yields

$$- \frac{dQ}{dx} = s C_b H \quad (10)$$

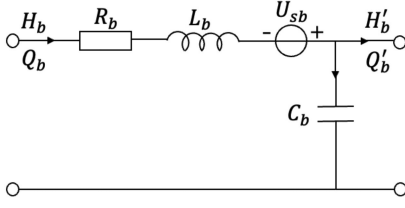


Fig. 2. Equivalent electric circuit for water flow movement in the pipe.

$$-\frac{dH}{dx} = (R_b + sL_b) Q - \frac{U_{sb}}{s} \quad (11)$$

Taking the derivative of (10) and (11) with respect to x , we have

$$\frac{d^2 Q}{dx^2} = sC_b \left((R_b + sL_b) Q - \frac{U_{sb}}{s} \right) = \gamma^2 Q - \zeta \quad (12)$$

$$\frac{d^2 H}{dx^2} = (R_b + sL_b) sC_b H = \gamma^2 H \quad (13)$$

where $\gamma^2 = (R_b + sL_b) sC_b$ and $\zeta = C_b U_{sb}$.

Solving (12) and (13) yields

$$Q_i = A_1^i(s) e^{-\gamma x} - A_2^i(s) e^{\gamma x} + \frac{\zeta}{\gamma^2} \quad (14)$$

$$H_i = B_1^i(s) e^{-\gamma x} + B_2^i(s) e^{\gamma x} \quad (15)$$

The index i denotes the i^{th} pipe.

Substituting (14) into (10) yields

$$H_i = \frac{A_1^i(s) \gamma e^{-\gamma x} + A_2^i(s) \gamma e^{\gamma x}}{sC_b} \quad (16)$$

Comparing (15) with (16), we have

$$B_1^i(s) = \frac{A_1^i(s) \gamma}{sC_b} \quad (17)$$

$$B_2^i(s) = \frac{A_2^i(s) \gamma}{sC_b} \quad (18)$$

Then the water flows and pressures along the pipe can be expressed as

$$\begin{cases} Q_i = A_1^i(s) e^{-\gamma x} - A_2^i(s) e^{\gamma x} + \frac{\zeta}{\gamma^2} \\ H_i = \frac{A_1^i(s) \gamma}{sC_b} e^{-\gamma x} + \frac{A_2^i(s) \gamma}{sC_b} e^{\gamma x} \end{cases} \quad (19)$$

Note that both the water flows and pressures are expressed as the algebraic equations in the Laplace domain.

Given two of the four variables Q_b , H_b , Q'_b and H'_b , it can be seen from Fig. 2 that $A_1^i(s)$ and $A_2^i(s)$ can be determined. Then the flows and pressures along the pipe can be obtained by taking the inverse Laplace transform of (19).

The pressures and the water flows along the pipe should satisfy the following constraints:

$$H_i(x, t) \geq H_{min} \quad (20)$$

$$Q_{min} \leq Q_i(x, t) \leq Q_{max} \quad (21)$$

H_{min} is the required minimum head pressure along the pipe. Q_{min} and Q_{max} denote the minimum and maximum volumes of the pipe.

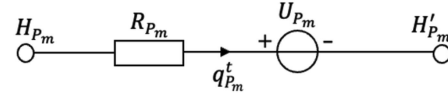


Fig. 3. Equivalent electric circuit for a fixed-speed pump.

$H_i(x, t)$ and $Q_i(x, t)$ denote the pressure and water flow at position x of the pipe i at time t , respectively. Since $Q_i(x, t)$ and $H_i(x, t)$ change continuously over the time and the position, the constraints from (20) to (21) can be replaced with the constraints from (22) to (25) for more close to the real-world applications. L_i denotes the length of pipe i , t_s the starting time and t_e the ending time. Mathematical proof of the constraint replacement is given in the Appendix.

$$H_i(0, t) \geq H_{min} \quad (22)$$

$$H_i(L_i, t) \geq H_{min} \quad (23)$$

$$Q_{min} \leq Q_i(0, t) \leq Q_{max} \quad (24)$$

$$Q_{min} \leq Q_i(L_i, t) \leq Q_{max} \quad (25)$$

The constraints (22) to (25) hold at $t = t_s$ or $t = t_e$.

C. Modeling of Water Pump, Valve and Tank

Here we describe the models of water pump, valve, and the tank used in the analogy model.

1) *Water Pump*: The pump is assumed to have fixed speed. The hydraulic model is given as

$$\text{Pump on : } \hat{h}_{P_m}^t = a_{P_m} (q_{P_m}^t)^2 + b_{P_m} q_{P_m}^t + c_{P_m} \quad (26)$$

$$q_{P_m}^{min} \leq q_{P_m}^t \leq q_{P_m}^{max} \quad (27)$$

$$\text{Pump off : } q_{P_m}^t = 0 \quad (28)$$

The relationship of pressure head gain and water flow in (26) can be rearranged as

$$\hat{h}_{P_m}^t = (2a_{P_m} q_{P_m,b}^t + b_{P_m}) q_{P_m}^t + c_{P_m} - a_{P_m} (q_{P_m,b}^t)^2 \quad (29)$$

The equivalent electric circuit for the pump is depicted in Fig. 3. The components in Fig. 3 are related to (29) as $H_{P_m} - H'_{P_m} = \hat{h}_{P_m}^t$, $R_{P_m} = 2a_{P_m} q_{P_m,b}^t + b_{P_m}$ and $U_{P_m} = c_{P_m} - a_{P_m} (q_{P_m,b}^t)^2$.

The electricity consumption of the water pump and that of the WDS can be calculated, respectively, as

$$P_{i,t}^{WP} = \kappa_1 (q_{P_i}^t)^3 + \kappa_2 (q_{P_i}^t)^2 + \kappa_3 q_{P_i}^t \quad (30)$$

$$P_{h,t}^{WDS} = \sum_{i=1}^{N_h^{WP}} P_{i,t}^{WP} \quad (31)$$

$$P_i \in \Delta_P \text{ and } h \in \Omega^{WDS}.$$

2) *Water Valve*: The pressure reducing valve is considered and the corresponding hydraulic model can be expressed as [31]

$$\text{Valve on : } \hat{h}_{V_m}^t = k_V (q_{V_m}^t)^2 \quad (32)$$

$$\text{Valve off : } q_{V_m}^t = 0 \quad (33)$$

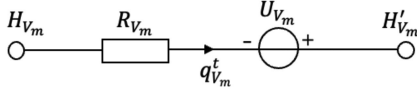


Fig. 4. Equivalent electric circuit for a pressure reducing valve.

where $\hat{h}_{V_m}^t$ is the pressure head gain and $q_{V_m}^t$ the flow of the valve m at time t . Equation (32) can be rewritten as

$$\hat{h}_{V_m}^t = 2k_V q_{V_m,b}^t q_{V_m}^t - k_V (q_{V_m,b}^t)^2 \quad (34)$$

The equivalent electric circuit for a pressure reducing valve is shown in Fig. 4, where $H_{V_m} - H'_{V_m} = \hat{h}_{V_m}^t$, $R_{V_m} = 2k_V q_{V_m,b}^t$ and $U_{V_m} = k_V (q_{V_m,b}^t)^2$.

The water flow in the valve should satisfy the constraint

$$q_{V_m}^{min} \leq q_{V_m}^t \leq q_{V_m}^{max} \quad (35)$$

3) *Water Tank*: The water tank can be modeled by the linear hydraulic constraints shown below

$$q_{J_m,t}^{in} + q_{J_m,t}^{out} = (h_{J_m}^{t+\Delta t} - h_{J_m}^t) A_{J_m}^{tank} \quad (36)$$

$$W_{J_m,t} = (1 - \mu_W) W_{J_m,t-\Delta t} + q_{J_m,t-\Delta t}^{in} + q_{J_m,t-\Delta t}^{out} \quad (37)$$

$$W_{J_m}^{min} \leq W_{J_m,t} \leq W_{J_m}^{max} \quad (38)$$

D. Discussions

1) More Explanation of Electrical-Water Analogy Method:

The proposed electrical-water analogy method consists of the following three main steps:

- Model the components of WDS as $f(V) = 0$, where $V = V_b + \Delta V$ is a generic variable used to represent the variables of WDS components and $f(\cdot)$ the hydraulic equation of WDS components. For the water pipe, V represents Q , the water flow in the pipe. For the fixed-speed pump, V is the variable $q_{P_m}^t$. For the valve, V refers to $q_{V_m}^t$. For the variable-speed pump, V refers to $q_{S_m}^t$. V also can be the variable $\omega_{S_m}^t$, the ratio of the actual pump speed to the nominal speed in the variable-speed pump. V_b is the baseline value of V , which can be calculated after solving the WDS optimization model without transfer delay consideration. ΔV is the difference between the optimized and baseline values of the variable.
- Rearrange and simplify the hydraulic model of WDS components using $V = V_b + \Delta V$.
- Make an analogy between the electric circuit and the components of WDS. Develop an equivalent electric circuit model for the WDS components.

Although the fixed-speed pump and the pressure reducing valve are considered, the proposed electrical-water analogy method can be applied to other kinds of components in the WDS. For example, the hydraulic model of the variable-speed pump can be formulated as

$$\hat{h}_{S_m}^t = a_{S_m} (q_{S_m}^t)^2 + b_{S_m} q_{S_m}^t \omega_{S_m}^t + c_{S_m} (\omega_{S_m}^t)^2 \quad (39)$$

The ratio of the actual pump speed to the nominal speed, $\omega_{S_m}^t$, should obey

$$0 \leq \omega_{S_m}^t \leq \hat{\omega}_{S_m} \quad (40)$$

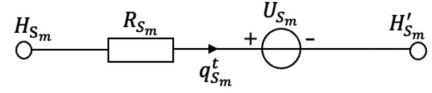


Fig. 5. Equivalent electric circuit for a variable-speed pump.

Substituting $q_{S_m}^t = q_{S_m,b}^t + \Delta q_{S_m}^t$ and $\omega_{S_m}^t = \omega_{S_m,b}^t + \Delta \omega_{S_m}^t$ into (39) yields

$$\begin{aligned} \hat{h}_{S_m}^t = & (2a_{S_m} q_{S_m,b}^t + b_{S_m} \omega_{S_m,b}^t) q_{S_m}^t \\ & + (b_{S_m} q_{S_m,b}^t \omega_{S_m}^t - b_{S_m} \omega_{S_m,b}^t q_{S_m}^t \\ & + 2c_{S_m} \omega_{S_m,b}^t \omega_{S_m}^t - a_{S_m} (q_{S_m,b}^t)^2 - c_{S_m} (\omega_{S_m,b}^t)^2) \end{aligned} \quad (41)$$

The equivalent electric circuit for the variable-speed pump is shown in Fig. 5, where components of the circuit are related to (41) via $H_{S_m} - H'_{S_m} = \hat{h}_{S_m}^t$, $R_{S_m} = 2a_{S_m} q_{S_m,b}^t + b_{S_m} \omega_{S_m,b}^t$, and $U_{S_m} = b_{S_m} q_{S_m,b}^t \omega_{S_m}^t - b_{S_m} \omega_{S_m,b}^t q_{S_m}^t + 2c_{S_m} \omega_{S_m,b}^t \omega_{S_m}^t - a_{S_m} (q_{S_m,b}^t)^2 - c_{S_m} (\omega_{S_m,b}^t)^2$.

2) *Water Transfer Delay on Wind Power Accommodation*: The electrical-water analogy model of the pipe in Fig. 2 contains inductors and capacitors, which are energy-storage elements from the perspective of circuit analysis. The amount of water storage in any segment of the pipe can be calculated by

$$W_{S_{x_1-x_2}}^{i,t_1-t_2} = \int_{t_1}^{t_2} (Q_i(x_1, t) - Q_i(x_2, t)) dt \quad (42)$$

where $W_{S_{x_1-x_2}}^{i,t_1-t_2}$ is the water stored between the positions x_1 and x_2 of pipe i during the time periods from t_1 to t_2 . With the transfer delays, the water pipe can be regarded as an auxiliary storage device for the WDS. It helps to deal with the variability of wind power. If power system faces a sudden increase of wind power, the pump in WDS will increase the electricity consumption and allow excess water be stored in the pipe to ensure the balance between supply and demand. If the wind power decreases suddenly, the electricity consumption of the pump in WDS will be lowered and the pipe will release the stored water to satisfy the water demand. Thus, power system can make full use of the energy flexibility of WDS to accommodate the abundant wind power.

Note that there may be concerns about the turbulent flow and the water quality. These concerns could be addressed by many measures. For example, the turbulent flow concern can be tackled by providing limitations on the speed of flow and hardening the water pipe [17], [32]. The water quality can be maintained by optimizing the chlorine dosage and the location of chlorine injection [33], [34].

Future research will be conducted to investigate the impact of water transfer delay on the turbulent flow and water quality for safe and economical operation of the WDS.

III. MODELING OF TRANSFER DELAY IN WDS

This section describes a graph-based searching method to model the water transmission process in the WDS network. The water in WDS is transported from the reservoir to the demand

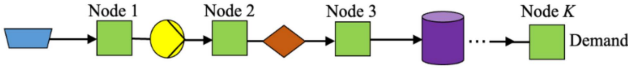


Fig. 6. Example of a one-way water transmission path. The water demand is at Node K .

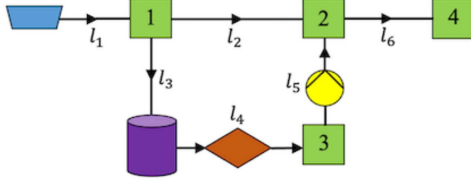


Fig. 7. Example of the water transmission network in WDS with cross nodes.

nodes through various paths. Two situations are considered. One is that the water transmission does not go through the cross nodes, another does. Details are given below.

A. Water Transmission in WDS Without Cross Nodes

The simplest path from the reservoir to the demand node is shown in Fig. 6, where the water is transported through a one-way path.

Suppose that the control variables such as the water flow in the pump and that of the valve, the outlet pressures of the pump and valve, the outlet flow of the tank and the reservoir, and the head pressure of the tank, are known. The head pressure of the reservoir is a fixed value determined by the geographical location of the reservoir. The water flows and pressures along the one-way path can be calculated by (19). The following constraint should be obeyed in the water transmission process

$$Q'_K = w_K \quad (43)$$

w_K denotes the water demand at node K and Q'_K refers to the calculated outlet flow of the pipe whose end node is node K . Constraint (43) states that the outlet flow of the pipe should be the same as the water demand at node K .

B. Water Transmission in WDS With Cross Nodes

In the real-world applications, the path from the reservoir to the demand node is not necessarily in the one-way form as shown in Fig. 6. There are many cross nodes at which multiple pipes join. Fig. 7 shows a water transmission network in WDS with cross nodes. There are two paths transporting water from the reservoir to the demand Node 4.

The following four steps are taken to deal with the water transmission with cross nodes.

1) *Search the Directed Paths*: There are various directed paths transporting water from the elements in $\mathcal{F}$, the set of reservoirs in the WDS, to the elements in $\mathcal{L}$, the set of demand nodes. The paths can be found using the graph theory-based algorithms [35].

2) *Find the Cross Node*: To find all the cross nodes in a WDS network, the node-branch incidence matrix is used

$$\begin{matrix} & l_1 & l_2 & \cdots & l_k & \cdots & l_M \\ \begin{matrix} n_1 \\ n_2 \\ \vdots \\ n_h \\ \vdots \\ n_N \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,k} & \cdots & a_{1,M} \\ a_{21} & a_{22} & \cdots & a_{2,k} & \cdots & a_{2,M} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{h,1} & a_{h,2} & \cdots & a_{h,k} & \cdots & a_{h,M} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{N,1} & a_{N,2} & \cdots & a_{N,k} & \cdots & a_{N,M} \end{bmatrix} \end{matrix} \quad (44)$$

The matrix has a size of $N \times M$, where N is the number of nodes and M the number of branches in the WDS. The element $a_{h,k}$ denotes the connection relationship between the node n_h and the branch l_k . The branch of a WDS denotes the pipe, pump or valve connecting two adjacent nodes. The direction of flow in the pipe and the on/off status of the pump and valve are determined before the cooperative optimization of the power system and WDS with transfer delay. The reason is presented in Section IV. The water flow in the branch with a pump or valve is in a fixed direction when the pump and valve are turned on. When they are off, there is no flow in the branch. Thus, the values of elements in the node-branch incidence matrix are fixed. If the flow in l_k starts from n_h , $a_{h,k} = 1$; if the flow ends at n_h , $a_{h,k} = -1$; if n_h is not an end of l_k or there is no flow in l_k , $a_{h,k} = 0$. The node h is a cross node if $\sum_{k=1}^M |a_{h,k}| > 2$. All the cross nodes in the WDS network can be found in this way. They are grouped together in the set $\mathcal{H}$.

3) *Group the Cross Node*: The cross nodes are sorted into several groups to determine the calculation sequence of the flows and pressures along the WDS network. Detailed process is given below.

The directed paths from the elements in $\mathcal{F}$ to the elements in $\mathcal{L}$ can be classified into two sets. The set $\mathcal{O}$ contains the directed paths that go through the cross nodes. The set $\mathcal{R}$ contains the directed paths that do not go through the cross nodes. Suppose that the cross node n_i is on the paths $p_1, p_2, \dots, p_J$ of set $\mathcal{O}$. The cross nodes that path p_j goes through are arranged in sequence. Let n_i be the r_j^{th} node and assigned to the $\max\{r_1, r_2, \dots, r_J\}$ group of the cross nodes. For example, in Fig. 7, there are two directed paths:

- p_1 : Reservoir $\rightarrow$ Node 1 $\rightarrow$ Tank $\rightarrow$ Valve $\rightarrow$ Node 3 $\rightarrow$ Pump $\rightarrow$ Node 2 $\rightarrow$ Node 4;
 p_2 : Reservoir $\rightarrow$ Node 1 $\rightarrow$ Node 2 $\rightarrow$ Node 4.

Nodes 1 and 2 are the cross nodes that p_1 goes through. Path p_2 also goes through Node 1 and Node 2. The reservoir is an element in $\mathcal{F}$ and the demand Node 4 is an element in $\mathcal{L}$. Node 1 ranks the first in both paths, i.e., $r_1 = 1$ and $r_2 = 1$. Thus, it is assigned to the group $\max\{11\} = 1$. Similarly, Node 2 is assigned to group $\max\{22\} = 2$.

The sequence of calculation is to calculate the flows and pressures from the reservoir to Node 1 in the first group, then those from the demand Node 4 to Node 2 in the second group. This is critical in the modeling of water transmission in WDS with cross nodes. Otherwise, the amount of flow in pipes l_2 and l_3 may not be able to obtain.

This situation can be explained as follows. The outlet flow of pipe l_1 and the head pressure of Node 1 can be obtained by (19), provided that the outlet flow and head pressure of the reservoir are known. It can be seen from Fig. 7 that there are two pipes, l_2 and l_3 , drawing water from Node 1. The one-way method presented in Section A cannot be applied to determine the amount of flows in pipes l_2 and l_3 . Note that the flows and pressures along the pipe can only be calculated when two of the four boundary conditions (the head pressures at both ends of the pipe, the inlet flow and the outlet flow) are given. For the cross Node 1, since only the calculated head pressure is known, the amount of flows in l_2 and l_3 cannot be calculated by (19).

With the cross node grouping described above, the head pressure of Node 2 can be calculated given the water demand and head pressure of Node 4. Then the water flow along l_2 can be obtained with the calculated head pressures of Nodes 1 and 2 by (19). Subtracting the inlet flow of l_2 from the outlet flow of l_1 yields the inlet flow of l_3 . Thus, by calculating the head pressures of the cross nodes first, the water flows and pressures in the whole WDS can be obtained.

4) *Calculation of Flow and Pressure in WDS*: Calculation of the flows and pressures in the WDS follows the steps below:

- Search all the directed paths from the elements in $\mathcal{F}/\mathcal{L}$ to each of the cross nodes in the first group/last group, and calculate the corresponding flows and head pressures along the paths using (19).
 - Add the elements in the first group/last group of the cross nodes to the set $\mathcal{F}/\mathcal{L}$. These nodes are then deleted from the cross node set $\mathcal{H}$.
 - Calculate the flows and head pressures along the paths using (19) for directed paths in the set $\mathcal{R}$. Based on the updated sets $\mathcal{F}$ and $\mathcal{L}$ in step b), update $\mathcal{R}$ by searching the directed paths from $\mathcal{F}$ to $\mathcal{L}$ that do not go through the cross nodes.
 - Go to step a) and continues until there are no more groups.
- The following constraints need to be met:

$$w_C = Q_{in,C} - Q_{out,C} \quad (45)$$

$$H_{C,1} = H_{C,2} = \dots = H_{C,n} \quad (46)$$

w_C denotes the water demand at the cross node (equals 0 if no water demand). $Q_{in,C}$ and $Q_{out,C}$ denote the total inlet flow and outlet flow of the cross node, respectively. Constraint (45) guarantees the mass balance at the cross node, while constraint (46) ensures that the calculated head pressures are the same at the cross node.

C. Constraints of WDS

Control variables of the WDS model include the water flows in the pump and in the valve, the outlet pressures of the pump and valve, the outlet flows of the tank and the reservoir, the head pressure of the tank, and the head pressures of the demand nodes. With the proposed graph-based searching method, the flows and pressures along the WDS are expressed by these variables considering water transfer delays. Delay constraints provide limitations on the values of the calculated water flows and pressures along the transfer delayed WDS. Delay constraints

of WDS include the constraints of WDS components and the constraints of water transmission process. The constraints of WDS components are expressed in (22) - (25), (27) - (31), and (33) - (38). Constraints of water transmission process include constraints (43), (45), and (46).

IV. COOPERATIVE OPERATION MODEL

Mathematical formulations of the cooperative operation model of the integrated electrical-water system are described below. The ultimate goal of the model is to minimize the total operation costs of the integrated system.

A. Cooperative Operation Model

The objective function of the cooperative operation model with transfer delay consideration is

$$C_S = \min \left(\sum_{i=1}^{N_g} \sum_{t=1}^T \left(c_i I_{i,t} + \sum_{l=1}^L K_{i,l} P_{i,t,l}^g \right) + \sum_{w=1}^{N_W} \sum_{t=1}^T \lambda P_{w,t}^{wc} + \sum_{h=1}^{N_{WDS}} \sum_{t=1}^T \lambda price_t P_{h,t}^{WDS} \right) \quad (47)$$

The production cost of the conventional unit can be expressed, using the piecewise linearization approximation [36], as a linear function

$$\sum_{i=1}^{N_g} \sum_{t=1}^T \sum_{l=1}^L K_{i,l} P_{i,t,l}^g.$$

Constraints of the cooperative operation model consists of the constraints of WDS and the power system. Constraints of WDS have been described in Section III-C. The constraints of the power system at any time instant t are listed below.

$$I_{i,t} - I_{i,t-1} = U_{i,t} - D_{i,t} \quad \forall i \in \Omega^T \quad (48)$$

$$U_{i,t} + D_{i,t} \leq 1 \quad \forall i \in \Omega^T \quad (49)$$

$$U_{i,t} + \sum_{l=1}^{TU_{i-1}} D_{i,t+l} \leq 1 \quad \forall i \in \Omega^T \quad (50)$$

$$D_{i,t} + \sum_{l=1}^{TD_{i-1}} U_{i,t+l} \leq 1 \quad \forall i \in \Omega^T \quad (51)$$

$$\left| \sum_{l=1}^L P_{i,t,l}^g - \sum_{l=1}^L P_{i,t-1,l}^g \right| \leq Ramp_i I_{i,t-1} + P_i^{gmin} (I_{i,t} - I_{i,t-1}) + P_i^{gmax} (1 - I_{i,t}) \quad \forall i \in \Omega^T \quad (52)$$

$$P_i^{gmin} I_{i,t} \leq \sum_{l=1}^L P_{i,t,l}^g \leq P_i^{gmax} I_{i,t} \quad \forall i \in \Omega^T \quad (53)$$

$$P_{i,t,l}^g \geq 0 \quad \forall i \in \Omega^T, l \in \Omega^L \quad (54)$$

$$\begin{aligned}
& \sum_{i \in \Omega_n^T} \sum_{l=1}^L P_{i,t,l}^g + \sum_{w \in \Omega_n^W} (P_{w,t}^{pre} - P_{w,t}^{wc}) = d_{n,t} \\
& + \sum_{h \in \Omega_n^{WDS}} P_{h,t}^{WDS} + \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{n,t} - \theta_{m,t}) \quad \forall n \in \Lambda^E
\end{aligned} \quad (55)$$

$$0 \leq P_{w,t}^{wc} \leq P_{w,t}^{pre} \quad \forall w \in \Omega^W \quad (56)$$

$$-PF_{n,m} \leq B_{n,m} (\theta_{n,t} - \theta_{m,t}) \leq PF_{n,m} \quad \forall (n, m) \in \Psi^E \quad (57)$$

$$\theta_{ref,t} = 0 \quad (58)$$

Constraints (48) and (49) restrict the status of the generating unit. Operation limitations of the unit are described by the constraints from (50) to (54). Constraint (55) is the nodal flow equation and constraint (56) gives the range of the wind power curtailment. Constraint (57) shows the direct current (DC) power flow limit of the line. Constraint (58) enforces the phase angle of the reference bus to zero.

The cooperative operation model considering water transfer delays is formulated as (59). It is a mixed-integer programming (MIP) problem and can be solved efficiently by commercially available solvers such as Gurobi.

$$\begin{cases} \min (46) \\ \text{s.t. } (22) - (25), (27) - (31), (33) - (38), (43) \\ (45) - (46), (48) - (58). \end{cases} \quad (59)$$

B. Two-Step Optimization Process

The cooperative optimization problem described above can be solved by a two-step process. The first step solves the optimization model with no transfer delays,

$$\begin{cases} \min (46) \\ \text{s.t. } (5), (20) - (21), (26) - (28), (30) - (33) \\ (35) - (38), (48) - (58) \\ w_h = \sum_{n:(n,h) \in \Delta} Q_{nh} - \sum_{m:(h,m) \in \Delta} Q_{hm} \quad \forall h \in \varpi, \end{cases} \quad (60)$$

where Q_{nh} and Q_{hm} denote the flows of pipes $n-h$ and $h-m$, respectively. The value of $\text{sign}(Q)$ and the on/off status of the pump and valve are obtained. Besides, Q_b , $q_{P_m,b}^t$, and $q_{V_m,b}^t$ are determined as the values of variables Q , $q_{P_m}^t$ and $q_{V_m}^t$.

In the second step, these fixed values are used as input parameters to calculate the water flows and pressures along the transfer delayed WDS with the proposed graph-based searching method. The calculated water flows and pressures are then used to solve the cooperative operation model (59) of power system and WDS to obtain the operational strategy that incorporates water transfer delays.

C. Fixed Values of Binary Variables

The values of $\text{sign}(Q)$ and the on/off status of the pump and valve serve as input parameters in the cooperative operation model (59) of power system and WDS. $\text{sign}(Q)$ denotes the direction of water flow in the pipe. Both $\text{sign}(Q)$ and the on/off status of the pump and valve are generally represented by binary

TABLE II
EQUIVALENT ELECTRICAL PARAMETERS OF THE PIPES IN WDS

	Pipe l_1	Pipe l_2	Pipe l_3	Pipe l_6
Resistance (Ω)	0.1908	0.2862	0.3339	0.3816
Inductance (H)	1.4443	1.4443	1.4443	1.4443
Capacitance (F)	6.92×10^{-7}	6.92×10^{-7}	6.92×10^{-7}	6.92×10^{-7}
Voltage source (V)	0.0191	0.0429	0.0584	0.0763

variables, which brings about hefty computational expense in the WDS optimization problem [18]. It may take hours to solve the optimization problem for a large-size WDS [27].

Since the output of wind power varies rapidly, the optimization model of WDS should be solved in minutes to cope with the wind power uncertainties. The long calculation time caused by the binary variables hinders the WDS from being fully used to optimize the power system operations with high penetration of wind power.

Thus, in this study, values of $\text{sign}(Q)$ and the on/off status of the pump and valve are calculated in advance. Then they are used as input parameters in the optimization problem. This relieves the concern of the computation time. Note that the fixed values of binary variables may affect the optimality of the obtained solution. In the future, research will be conducted to improve the approach for determining the values of these binary variables and make the obtained solution as close to the optimal solution as possible.

V. CASE STUDY

The proposed electrical-water analogy model is evaluated to demonstrate that more wind power can be accommodated into the power system and the operation cost of the integrated electrical-water system can be reduced considering the water transfer delays in WDS. Experiments were performed on a small-scale illustrative test system and a large-scale test system. The time horizon is 24 hours and the study resolution is 15 minutes. This corresponds to 96 time steps. The tests were carried out by a PC with Intel Core i7-9700 and Gurobi 9.0.1 on Python 3.7.

A. Illustrative Test System

First, a relatively small test system is used to illustrate the performance of the proposed electrical-water analogy model in detail. It contains a 6-bus power system, shown in Fig. 1, and the WDS with 4 pipes, shown in Fig. 7. There are 3 load buses in the 6-bus power system and each load bus is connected with an identical WDS. Data of the 6-bus power system can be found in [27].

The electrical-water analogy parameters of the pipes in the WDS are given in Table II. The required minimum head pressure along the pipe is set as 20 m. The diameter, length and volume of each pipe are listed in Table III. The volume of the pipe is calculated by $\pi \times (\frac{\text{Diameter}}{2})^2 \times \text{length}$. Parameters of the pump and the valve are given in Table IV. The wind power output and water demand curves are shown in Fig. 8. Four scenarios are considered. Scenario *a* reflects the daily wind power and

TABLE III
DIMENSION PARAMETERS OF THE PIPES IN WDS

	Pipe l_1	Pipe l_2	Pipe l_3	Pipe l_6
Diameter (m)	0.2	0.2	0.2	0.2
Length (m)	7,000	8,047	8,300	8,500
Volume (m ³)	219.80	252.68	260.62	266.90

TABLE IV
PARAMETER VALUES IN HYDRAULIC EQUATIONS OF PUMP AND VALVE

Parameter	a_{p_m}	b_{p_m}	c_{p_m}	k_v
Value	-1.0941×10^{-4}	0.5152	40	6.13

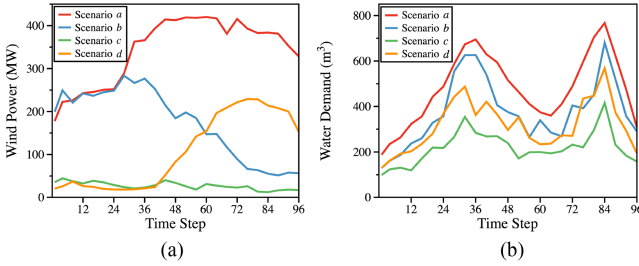


Fig. 8. Daily wind power output and water demand for various scenarios. Data are collected every 15 minutes for a 24-hour period.

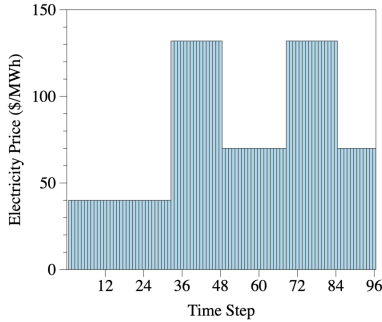


Fig. 9. Electricity price.

water demand for the months from March to May, Scenario b from June to August, Scenario c from September to November, and Scenario d from December to February. The penalty for wind power curtailment is set as 1000 \$/MWh. Fig. 9 shows the electricity price used in [27]. To encourage the use of wind power, the electricity price for WDS is set to be 0.5 times of the original price, if WDS agrees to cooperate with the power system.

The water transfer delay is defined as the amount of time to transport water from the source to the user. The delay is generally affected by two factors. One is the value of inductance and capacitance in Table II. Another factor is the size of WDS. Larger inductance and capacitance imply that more time is needed to transport water from the beginning to the end of a pipe. On the other hand, a large-size WDS means the distance between the water source and the user expands, which increases the transfer delay.

TABLE V
WIND POWER CURTAILMENT IN DIFFERENT SITUATIONS FOR THE ILLUSTRATIVE TEST SYSTEM

Situation	Scenario	Wind power curtailment (MWh)
1	a	1,353.78
	b	1,005.60
	c	60.56
	d	593.54
2	a	676.21
	b	514.90
	c	0
	d	282.05
3	a	387.36
	b	209.67
	c	0
	d	121.58

1) *Wind Power Accommodation*: Three situations are considered to verify the benefits of considering the transfer delay for wind power accommodation.

Situation 1: Consider the power system alone.

Situation 2: Consider power system and WDS without the transfer delays of water.

Situation 3: Consider power system and WDS with the transfer delays of water.

Note that the cooperative operation model of Situation 2 has been considered in [12], [18] and [27]. In each situation, four daily scenarios of the wind power output and water demand are considered for seasonal impact.

The amount of wind power curtailment for each scenario under three situations is shown in Table V. It is observed from Table V that the amount of wind power curtailment is reduced significantly after the cooperative operation between power system and WDS. The wind power curtailments of Scenario c in Situations 2 and 3 are zero, because the wind power output remains at a low level, as evidenced in Fig. 8(a). Thus, wind power variations can be compensated by regulating the electricity consumption of WDS. It also can be seen from Table V that the amounts of wind power curtailment in Scenarios a , b and d under Situation 3 are reduced by about 50% compared with that in Situation 2. This shows the merits of considering water transfer delays for wind power accommodation. Since Situation 2 does not consider the water transfer delay, its water pipe does not offer the storage capability, which leads to more curtailment. Details are described below.

The transfer delay enables the water pipe to act as an auxiliary storage device for the WDS. Thus, in Situation 3, WDS is able to use both the tank and pipe for water storage and release. In Situation 2, WDS only can use the tank for water storage and release. The storage capability of pipe is elaborated by Fig. 10, which shows the inlet and outlet flows of pipe l_6 . It can be seen from Fig. 10 that, when the inlet flow increases, it takes about 30 minutes for the outlet flow to increase. Similar delay occurs when the inlet flow decreases. Thus, the water transfer delay is 30 minutes in the pipe. It also is observed from Fig. 10 that the pipe stores the excess water when the inlet flow outweighs

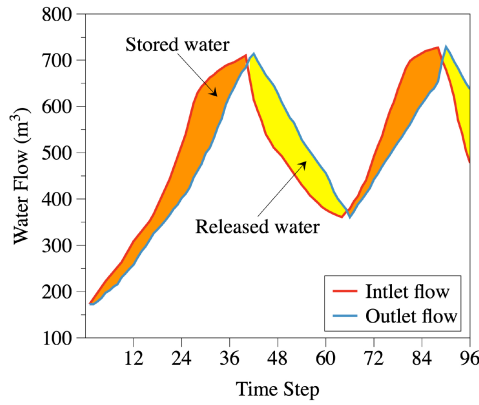
Fig. 10. Inlet and outlet flows of pipe l_6 .

TABLE VI
STORAGE CAPACITY OF WATER PIPE AND TANK IN THE ILLUSTRATIVE TEST SYSTEM

Delay	Scenario	Pipe Capacity (m^3)	Tank Capacity (m^3)
No	<i>a</i>	0	4,854.05
	<i>b</i>	0	4,328.10
	<i>c</i>	0	3,534.83
	<i>d</i>	0	3,916.27
Yes	<i>a</i>	4,331.76	4,048.98
	<i>b</i>	3,745.79	3,246.07
	<i>c</i>	2,682.14	2,649.12
	<i>d</i>	3,020.81	3,190.86

the outlet flow. The stored water is released when the inlet flow is less. This operation helps the balance of water supply and demand.

Next, we investigate the storage capacity of the pipe and the tank. Table VI shows the capacity of the pipe and the water tank at various scenarios. The capacity here refers to the total amount of daily water storage, which varies at different scenarios to accommodate the wind power with different output profiles. The maximum threshold of the water storage in the tank is 5000 m^3 satisfying the technical limitations. The dimensions of pipes are presented in Table III. When the transfer delay is not considered, the pipe is not for storage. It is observed from Table VI that, considering the transfer delay, the pipe storage capacity can be up to 95% to 115% of the tank capacity. This brings about the total amount of stored water to be about 1.51 to 1.73 times that of using only the tank. As a result, the consideration of transfer delay improves the water storage capability of the WDS. This allows more water to be pumped into the WDS during the period of high wind power output.

The storage capability of pipe allows the WDS to shift the electricity consumption profile more flexibly in Situation 3, as shown in Fig. 11. With the transfer delay, WDS can increase the electricity consumption to store the excess water in the pipes and tanks when the wind power resource is abundant, and release the stored water for water balance when the wind power output is low. This adds more flexibility in the cooperative operation of the power system and WDS and leads to less wind power curtailment than that of Situation 2.

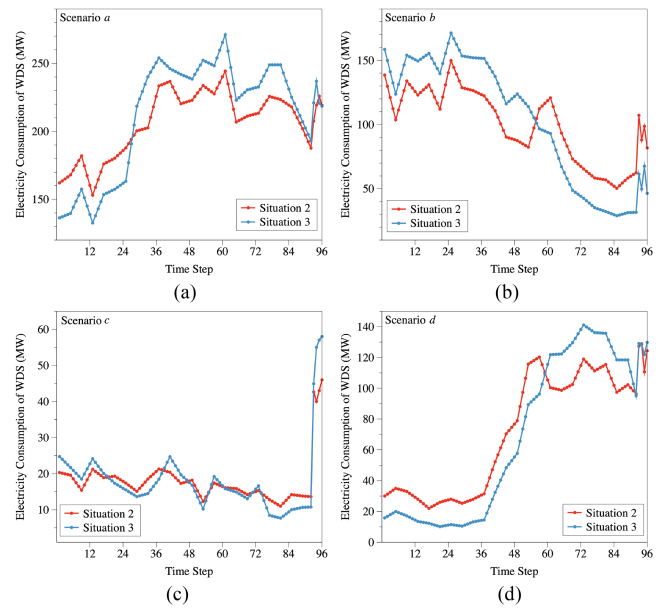


Fig. 11. Electricity consumption of WDS not considering the transfer delay (Situation 2) and considering the delay (Situation 3).

TABLE VII
IMPACT OF THE CROSS-SECTIONAL AREA OF THE PIPE ON WIND POWER ACCOMMODATION

A (cm^2)	Transfer delay (min)	Scenario	Wind power curtailment (MWh)
78.5	60	<i>a</i>	366.19
		<i>b</i>	187.68
		<i>c</i>	0
		<i>d</i>	108.05
176.5	45	<i>a</i>	374.36
		<i>b</i>	190.52
		<i>c</i>	0
		<i>d</i>	110.13
314	30	<i>a</i>	387.36
		<i>b</i>	209.67
		<i>c</i>	0
		<i>d</i>	121.58
490.5	15	<i>a</i>	401.61
		<i>b</i>	251.74
		<i>c</i>	0
		<i>d</i>	152.31
706.5	8	<i>a</i>	445.80
		<i>b</i>	292.29
		<i>c</i>	0
		<i>d</i>	186.32

Water transfer delay is also affected by the cross-sectional area of the pipe. Larger cross-sectional area usually allows more water to flow through, and in turn, less transfer delay. Table VII shows the impact of the size of pipe cross-sectional area on the transfer delay and wind power curtailment. Five cross-sectional areas are considered. It can be seen from Table VII that larger cross section area does lower the transfer delay. Smaller transfer delay implies higher wind power curtailment, which may bring about extra penalty cost for power system.

Smaller cross section areas lead to higher water losses due to the increased water flow. Higher water losses mean that the water pumps need to consume more electricity to increase the

TABLE VIII
IMPACT OF THE CROSS-SECTIONAL AREA OF THE PIPE ON ELECTRICITY BILL OF WDS AND PENALTY FOR WIND POWER CURTAILMENT

A (cm ²)	Electricity bill of WDS (\$)	Penalty for wind power curtailment (\$)	Total cost (\$)
78.5	228,023	165,480	393,503
176.5	186,832	168,752	355,584
314	121,194	179,652	300,846
490.5	104,621	201,415	306,036
706.5	87,652	231,102	318,754

TABLE IX
OPERATION COSTS OF THE ILLUSTRATIVE TEST SYSTEM

System type	Delay	Scenario	Independent Operation	Coordinated Operation
Power system	No	a	\$1,507,925	\$714,637
		b	\$1,254,738	\$560,718
		c	\$74,985	\$24,887
		d	\$731,782	\$308,206
	Yes	a	\$1,507,925	\$408,616
		b	\$1,254,738	\$220,704
		c	\$74,985	\$18,063
		d	\$731,782	\$132,608
WDS	No	a	\$391,710	\$248,666
		b	\$182,364	\$107,652
		c	\$29,185	\$18,720
		d	\$120,419	\$91,401
	Yes	a	\$307,324	\$263,213
		b	\$153,483	\$108,654
		c	\$25,681	\$20,615
		d	\$103,817	\$92,296

pressure difference. Thus, the electricity bill of WDS increases. Table VIII presents the impact of cross-sectional area on the electricity bill of WDS and the penalty for wind power curtailment. The second and the third columns indicate the average value of four scenarios. It is observed from Table VIII that the electricity bill of WDS is lowered as A increases, whereas the penalty for wind power curtailment increases as the size of A increases. It can be seen from Table VIII that the smallest total cost occurs at the cross-sectional area of 314 cm². Thus, this study suggests that a pipe with cross sectional size of 314 cm² provides a good tradeoff between water loss and wind power curtailment.

2) *Operation Cost*: Two operation modes of the power system and WDS are considered. One is the independent operation, another is the coordinated operation.

Table IX compares the operation costs of these two operations in the illustrative test system. The second column indicates whether the water transfer delay is considered in the WDS model. It can be seen from Table IX that the operation cost of each energy system in each scenario has decreased significantly after the coordination, by about 11% to 82%. The substantial cost reduction would encourage the energy systems to cooperate with each other instead of working independently. Table IX also reveals that, after the coordination, the operation cost of WDS considering transfer delay is larger than that not considering the transfer delay. This is because the transfer delay enables WDS to consume more electricity for wind power accommodation

TABLE X
TOTAL OPERATION COST OF INTEGRATED ELECTRICAL-WATER SYSTEM AFTER COORDINATION IN THE ILLUSTRATIVE TEST SYSTEM

Delay	Scenario	Total cost (\$)
No	a	963,303
	b	668,370
	c	43,607
	d	399,607
Yes	a	671,829
	b	329,358
	c	38,678
	d	224,904

TABLE XI
OPTIMIZATION RESULTS OF THE PROPOSED ELECTRICAL-WATER ANALOGY METHOD AND THE FD METHOD IN THE ILLUSTRATIVE TEST SYSTEM

Method	Scenario	Total cost (\$)	Time (sec)
Proposed	a	671,829	39.97
	b	329,358	38.44
	c	38,678	41.52
	d	224,904	36.83
FD (1 minute, 100 meters)	a	678,027	564.79
	b	334,108	568.10
	c	39,705	681.72
	d	230,016	521.34
FD (30 seconds, 50 meters)	a	671,829	9,543.50
	b	329,358	7,639.38
	c	38,678	10,367.62
	d	224,904	7,169.72

and the electricity bill of WDS increases. Nevertheless, total operation cost of the integrated electrical-water system has been reduced significantly due to less wind power curtailment, as evidenced by Table X. This validates the economic merit of incorporating the water transfer delay into the optimization model. In the future, the optimal pricing scheme for WDS will be investigated to reduce the electricity bill under the transfer delay situation.

3) *Performance of the Electrical-Water Analogy Method*: Next, we show that the proposed graph-based electrical-water analogy method significantly lowers the computational cost, while maintains almost the same operation cost, in comparison to the FD method. The proposed method transforms the PDEs into the algebraic equations in the Laplace domain, which facilitates the computations in obtaining the results. Table XI displays the computational time and the operation cost at various scenarios for the illustrative test system. Two cases are considered for the FD method in Table XI. The 1st case sets the time at 1 minute and the space steps at 100 meters. The 2nd case is set at 30 seconds and 50 meters. It is observed from Table XI that the proposed method returns a total cost lower than the FD method in the 1st case. The accuracy of the FD method depends on the number of segments of the time and space. More segments lead to more accurate results. The total cost calculated by FD is the same as that of the proposed method when the time and space steps follow the settings of the 2nd case. However, the computation time of the FD method increases dramatically as the number of segments increases. In comparison, computational cost of the proposed method is significantly lower.

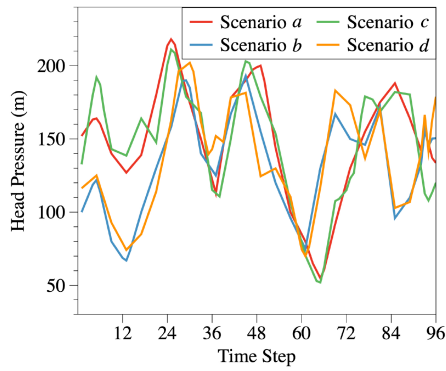


Fig. 12. Head pressure at Node 4, the end of WDS.

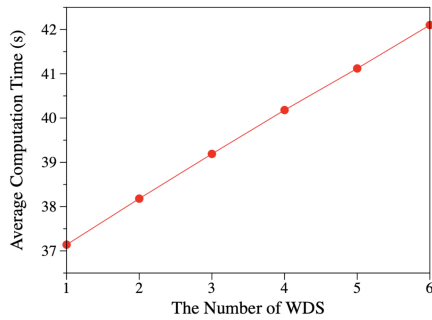


Fig. 13. Impact of the number of WDS on the computation time of the illustrative system.

The head pressure of Node 4, which is at the end of the WDS, is the lowest. Fig. 12 shows the variations of the calculated head pressure at Node 4, the end node of WDS. The findings demonstrate that the proposed method can satisfy the minimum pressure requirement of 20 m.

The scalability of the proposed electrical-water analogy method is evaluated by increasing the number of WDS connected to the power system. The number of the test WDS is set to vary from 1 to 6. Fig. 13 shows the impact of the number of WDS on the consumed time. The computation time is averaged out of four scenarios. It can be seen from Fig. 13 that as the number of WDS increases, the average consumed time to solve the cooperative optimization problem grows linearly. This indicates the linear scalability of the proposed method.

B. Large-scale Test System

This section evaluates the performance of the proposed electrical-water analogy method and demonstrates the benefits of water transfer delay consideration on a large-scale system. The test system consists of the IEEE-RTS 24 power system and the WDS with 40 pipes [18]. Each load bus in the power system is connected with a WDS. The transfer delay is around 200 minutes.

1) *Wind Power Accommodation*: Evaluation of wind power accommodation was carried out using the three situations considered in the illustrative test system. Table XII shows the wind power curtailments at different scenarios. Situation 3 considers

TABLE XII
AMOUNT OF WIND POWER CURTAILMENT IN DIFFERENT SITUATIONS IN THE LARGE-SCALE TEST SYSTEM

Situation	Scenario	Wind power curtailment (MWh)
1	<i>a</i>	3,426.95
	<i>b</i>	2,587.36
	<i>c</i>	179.39
	<i>d</i>	1,537.98
2	<i>a</i>	1,220.75
	<i>b</i>	634.24
	<i>c</i>	0
	<i>d</i>	252.48
3	<i>a</i>	983.17
	<i>b</i>	437.41
	<i>c</i>	0
	<i>d</i>	174.94

TABLE XIII
STORAGE CAPACITY OF WATER PIPE AND TANK IN THE LARGE-SCALE TEST SYSTEM

Delay	Scenario	Pipe Capacity (m ³)	Tank Capacity (m ³)
No	<i>a</i>	0	13,267.76
	<i>b</i>	0	11,603.75
	<i>c</i>	0	9,980.26
	<i>d</i>	0	10,450.48
Yes	<i>a</i>	12,753.70	11,405.71
	<i>b</i>	9,960.15	10,316.34
	<i>c</i>	7,136.78	8,731.89
	<i>d</i>	8,382.26	9,109.63

the water transfer delay and cooperative operation, while Situation 2 considers only the cooperative operation. It can be seen from Table XII that, with the cooperative operation of the power system and WDS, the wind power curtailment can be reduced by more than 60%. The consideration of water transfer delay enables the reduction of another 20% to 30% of wind power curtailment.

The storage capacity of the pipe and the tank in the large-scale system is investigated in the following. Table XIII shows the total amounts of daily stored water in the pipes and in the water tank. The maximum water storage in the tank is 15000 m³. The water volume in each pipe should be less than 500 m³, according to the physical limitations. Since the transfer delay prevents the water from being transported into the tank for storage instantly, considerable amounts of the water are stored in the pipes during the transportation process. This makes the storage capacity of the pipe be up to 82% to 112% of the tank capacity with the transfer delay consideration. And the total amounts of stored water in the pipes and tank are about 1.59 to 1.82 times of that not considering the delay.

2) *Operation Cost*: Table XIV and Table XV compare the operation cost of energy systems between the independent operation mode and the coordinated operation mode. It can be seen that the operation costs of both energy systems have decreased by about 10% to 82% after the coordination, which validates the benefit of system coordination in reducing operation cost. In addition, the total cost of the integrated system is reduced greatly considering the water transfer delay.

TABLE XIV
OPERATION COSTS OF THE LARGE-SCALE TEST SYSTEM

System type	Delay	Scenario	Independent Operation	Coordinated Operation
Power system	No	a	\$2,045,109	\$972,673
		b	\$1,700,193	\$745,533
		c	\$100,861	\$35,049
		d	\$1,001,724	\$445,353
	Yes	a	\$2,045,109	\$540,928
		b	\$1,700,193	\$318,465
		c	\$100,861	\$25,898
		d	\$1,001,724	\$181,366
WDS	No	a	\$689,046	\$359,938
		b	\$259,982	\$151,471
		c	\$46,491	\$27,310
		d	\$217,900	\$133,474
	Yes	a	\$672,737	\$610,475
		b	\$215,645	\$155,107
		c	\$38,154	\$29,343
		d	\$172,059	\$140,855

TABLE XV
TOTAL OPERATION COST OF INTEGRATED ELECTRICAL-WATER SYSTEM AFTER COORDINATION IN THE LARGE-SCALE TEST SYSTEM

Delay	Scenario	Total cost (\$)
No	a	1,332,611
	b	897,004
	c	62,359
	d	578,827
Yes	a	1,151,403
	b	473,572
	c	55,241
	d	322,221

TABLE XVI
OPTIMIZATION RESULTS OF THE PROPOSED ELECTRICAL-WATER ANALOGY METHOD AND THE FD METHOD IN THE LARGE-SCALE TEST SYSTEM

Method	Scenario	Total cost (\$)	Time (sec)
Proposed	a	1,151,403	45.52
	b	473,572	43.18
	c	55,241	47.26
	d	322,221	39.80
FD	a	1,149,475	1,311.41
	b	472,730	1,215.63
	c	53,393	1,465.82
	d	321,082	1,097.05

3) *Performance of the Electrical-Water Analogy Method:* Performance comparison between the proposed method and the FD method was conducted. The time and space steps were set at 1 minute and 100 meters, respectively. Results on the operation cost and the computation time at various scenarios are shown in Table XVI. Moreover, the scalability of the proposed algorithm is validated by varying the number of WDS from 15 to 22, as shown in Fig. 14. From Table XVI and Fig. 14, it is concluded that the proposed method not only produces almost the same operation costs as the FD method, but also significantly reduces the computation time. It takes roughly 44 seconds on average for the proposed method to obtain the result, whereas the FD method takes 1272 seconds. Besides, the scalability of the proposed algorithm grows linearly. The longest optimization time occurs when the number of WDS is 22, whereas it only takes 88 seconds

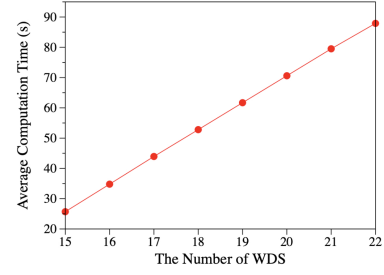


Fig. 14. Impact of the number of WDS on the computation time of the large-scale system.

to obtain the optimal solution which is short enough for practical use.

VI. CONCLUSION

This paper has presented a WDS optimization model to improve the operation of integrated electrical-water system considering the water transfer delay. The water transmission process in the WDS is modeled by a graph-based searching scheme. The proposed method improves the performance of WDS in power system by further reducing the wind power curtailment and the operation cost. Moreover, it efficiently solves the PDEs of WDS to lower the computation cost dramatically.

Results from case studies have demonstrated that the cooperative operation of power system and WDS considering the water transfer delays further improves the reduction of wind power curtailment. Operation costs of the two energy systems have been reduced considerably after coordinating with each other. Moreover, the proposed method significantly lowers the computational cost compared with the FD method and the scalability of the proposed method is linear.

In addition to promoting the wind power accommodation and reducing the operation cost, the proposed model can be extended to other electrical-water systems, such as the energy flexibility analysis [12], the bidding strategy in the electricity market [14], and the tank size optimization [18]. Future work will investigate the impact of transfer delay on the above applications.

VI. APPENDIX

Theorem 1 confirms that the constraints from (20) to (21) can be simplified as the constraints from (22) to (25) without affecting the tightness. Prior to the detailed proof of Theorem 1, Lemma 1 and Lemma 2 are needed.

Lemma 1: The inverse Laplace transform of $e^{-\gamma x}$ and $e^{\gamma x}$ are

$$\begin{cases} L^{-1}(e^{-\gamma x}) = e^{-\frac{R_b}{2L_b}t} \delta(t - \sqrt{L_b C_b}x) \\ + e^{-\frac{R_b}{2L_b}t} \frac{R_b \sqrt{L_b C_b}x}{\sqrt{t^2 - L_b C_b x^2}} I_1\left(\frac{R_b}{2L_b} \sqrt{t^2 - L_b C_b x^2}\right) u(t - \sqrt{L_b C_b}x) \\ L^{-1}(e^{\gamma x}) = \frac{-R_b e^{-R_b t/2L_b}}{L_b \sqrt{x^2 L_b C_b - t^2}} x \sqrt{L_b C_b} \mathcal{T}_1\left(\frac{R_b}{2L_b} \sqrt{x^2 L_b C_b - t^2}\right) \\ (u(t) - u(t - \sqrt{L_b C_b}x)), \end{cases}$$

where $\delta(t)$ denotes the unit impulse function, I_1 is the modified Bessel function of the first kind, $u(t)$ the unit step function, and J_1 the Bessel function of the first kind.

Proof of Lemma 1: Define $G_1(s)$ and $G_2(s)$ as

$$\begin{cases} G_1(s) = e^{-x\sqrt{L_b C_b}} \sqrt{s^2 - (R_b)^2 / 4(L_b)^2} \\ G_2(s) = e^{x\sqrt{L_b C_b}} \sqrt{s^2 - (R_b)^2 / 4(L_b)^2} \end{cases} \quad (A1)$$

With the shifting property of Laplace transform, $L^{-1}(e^{-\gamma x})$ and $L^{-1}(e^{\gamma x})$ can be transformed into

$$\begin{cases} L^{-1}(e^{-\gamma x}) = e^{-R_b t / 2L_b} L^{-1}(G_1(s)) \\ L^{-1}(e^{\gamma x}) = e^{-R_b t / 2L_b} L^{-1}(G_2(s)) \end{cases} \quad (A2)$$

According to [37], the inverse Laplace transform of $G_1(s)$ is given as

$$L^{-1}(G_1(s)) = \delta(t - \sqrt{L_b C_b} x) + \frac{\frac{R_b \sqrt{L_b C_b} x}{2L_b}}{\sqrt{t^2 - L_b C_b x^2}} I_1(R_b \sqrt{t^2 - L_b C_b x^2} / 2L_b) u(t - \sqrt{L_b C_b} x). \quad (A3)$$

For the inverse Laplace transform of $G_2(s)$, the following contour integral is required:

$$\oint_C e^{\zeta \sqrt{s^2 - \kappa^2}} e^{tz} dz \quad (A4)$$

where $\zeta = x\sqrt{L_b C_b}$ and $\kappa = R_b / (2L_b)$.

By Cauchy's theorem, the contour integral of (A4) equals zero. The contour integral (A4) can be expressed as.

$$\begin{aligned} & \int_{c-i\sqrt{R^2-c^2}}^{c+i\sqrt{R^2-c^2}} e^{\zeta \sqrt{s^2 - \kappa^2}} e^{ts} ds + iR \int_{\pi/2 - \arcsin(\frac{c}{R})}^{\pi} e^{i\theta} e^{\zeta \sqrt{(Re^{i\theta})^2 - \kappa^2}} e^{Rte^{i\theta}} d\theta - \int_R^{\kappa} e^{\zeta \sqrt{x^2 - \kappa^2}} e^{-tx} dx \\ & - \int_{\kappa}^{-\kappa} e^{-i\zeta \sqrt{\kappa^2 - x^2}} e^{-tx} dx \\ & - \int_{-\kappa}^{\kappa} e^{i\zeta \sqrt{\kappa^2 - x^2}} e^{-tx} dx - \int_{\kappa}^R e^{\zeta \sqrt{x^2 - \kappa^2}} e^{-tx} dx + iR \\ & \int_{\pi}^{3\pi/2 + \arcsin(\frac{c}{R})} e^{i\theta} e^{\zeta \sqrt{(Re^{i\theta})^2 - \kappa^2}} e^{Rte^{i\theta}} d\theta \end{aligned} \quad (A5)$$

where R is the radius of the contour and the third and sixth integrals cancel. Further, the second and seventh integrals can be combined as:

$$\begin{aligned} & iR \int_{\pi/2 - \arcsin(\frac{c}{R})}^{3\pi/2 + \arcsin(\frac{c}{R})} e^{i\theta} e^{\zeta Re^{i\theta} \sqrt{1 - \frac{\kappa^2}{(Re^{i\theta})^2}}} e^{Rte^{i\theta}} d\theta \\ d\theta &= \lim_{R \rightarrow \infty} iR \int_{\pi/2}^{3\pi/2} e^{i\theta} e^{R(\cos \theta + i \sin \theta)(\zeta + t)} d\theta \\ &= \lim_{R \rightarrow \infty} iR \int_{\pi/2}^{3\pi/2} e^{i\theta} e^{R(\zeta + t) \cos \theta} (\cos(R(\zeta + t) \sin \theta) \\ &+ i \sin(R(\zeta + t) \sin \theta)) d\theta = 0 \end{aligned} \quad (A6)$$

As $R \rightarrow \infty$, the inverse Laplace transform of $G_2(s)$ is transformed into (A7) by the Cauchy's theorem:

$$\frac{1}{i2\pi} \int_{c-i\sqrt{R^2-c^2}}^{c+i\sqrt{R^2-c^2}} e^{\zeta \sqrt{s^2 - \kappa^2}} e^{ts} ds = \frac{1}{i2\pi} \left(\int_{-\kappa}^{\kappa} e^{i\zeta \sqrt{\kappa^2 - x^2}} \right.$$

$$\left. - e^{-i\zeta \sqrt{\kappa^2 - x^2}} \right) e^{-tx} dx = \frac{1}{\pi} \int_{-\kappa}^{\kappa} \sin \zeta \sqrt{\kappa^2 - x^2} e^{-tx} dx. \quad (A7)$$

Substituting $x = \kappa \cos \frac{\theta}{2}$ into (A7), we have

$$\begin{aligned} & \frac{1}{\pi} \int_{-\kappa}^{\kappa} \sin \zeta \sqrt{\kappa^2 - x^2} e^{-tx} dx \\ &= -\frac{\kappa}{2\pi} \int_{-\pi}^{\pi} \sin \frac{\theta}{2} \sin \left(\zeta \kappa \sin \frac{\theta}{2} \right) e^{-t\kappa \cos \frac{\theta}{2}} d\theta \\ &= -\frac{\kappa}{2\pi} \text{Im} \int_{-\pi}^{\pi} \sin \frac{\theta}{2} e^{i\zeta \kappa \sin \frac{\theta}{2}} e^{-t\kappa \cos \frac{\theta}{2}} d\theta \end{aligned} \quad (A8)$$

The integral on the right-hand side of (A8) can be arranged as

$$\begin{aligned} & \int_{-\pi}^{\pi} \sin \frac{\theta}{2} e^{i\zeta \kappa \sin \frac{\theta}{2}} e^{-t\kappa \cos \frac{\theta}{2}} d\theta \\ &= -i2 \oint_{|z|=1} \frac{z - z^{-1}}{i2} e^{\frac{\zeta \kappa (z - z^{-1})}{2}} e^{-\frac{t\kappa (z + z^{-1})}{2}} \frac{dz}{z} \\ &= - \oint_{|z|=1} (z - z^{-1}) e^{\frac{\zeta \kappa - t\kappa}{2} z - \frac{\zeta \kappa + t\kappa}{2} z^{-1}} \frac{dz}{z} \end{aligned} \quad (A9)$$

The contour integral is equal to $i2\pi$ times the sum of the residues of the poles inside of the unit circle. In this case, the integrand has an essential singularity at $z = 0$. To compute the residue at $z = 0$, the exponential is expanded in a Laurent series and only the coefficient of z^{-1} in the expansion needs to be determined. In short, we wish to compute the coefficient of z^0 in the Laurent expansion of

$$(z - z^{-1}) e^{az - bz^{-1}} \quad (A10)$$

where $a = \frac{\zeta \kappa - t\kappa}{2}$ and $b = \frac{\zeta \kappa + t\kappa}{2}$. This expansion is

$$\begin{aligned} (z - z^{-1}) e^{az - bz^{-1}} &= (z - z^{-1}) \sum_{k=0}^{\infty} \frac{1}{k!} (az - bz^{-1})^k \\ &= (z - z^{-1}) \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{m=0}^k \binom{k}{m} (az)^m (-bz^{-1})^{k-m} \\ &= (z - z^{-1}) \sum_{k=0}^{\infty} b^k \sum_{m=0}^k (-1)^{k-m} \frac{1}{m! (k-m)!} \left(\frac{a}{b}\right)^m z^{2m-k} \end{aligned} \quad (A11)$$

Note that only the odd values of k will lead to a nonzero coefficient of z^0 . The last sum is then rewritten as

$$\begin{aligned} (z - z^{-1}) e^{az - bz^{-1}} &= (z - z^{-1}) \sum_{k=0}^{\infty} b^{2k+1} \sum_{m=0}^k (-1)^{2k+1-m} \\ &\frac{1}{m! (2k+1-m)!} \left(\frac{a}{b}\right)^m z^{2m-2k-1} \end{aligned} \quad (A12)$$

Nonzero coefficients of z^0 appear when $m = k$ or $m = k + 1$. By the residue theorem, the integral above is equal to

$$\int_{-\pi}^{\pi} \sin \frac{\theta}{2} e^{i\zeta \kappa \sin \frac{\theta}{2}} e^{-t\kappa \cos \frac{\theta}{2}} d\theta = -i2\pi \left(\sum_{k=0}^{\infty} (-1)^{k+1} \right.$$

$$\begin{aligned}
& \frac{a^k b^{k+1}}{k! (k+1)!} - \sum_{k=0}^{\infty} (-1)^k \frac{a^{k+1} b^k}{k! (k+1)!} \Bigg) \\
&= \frac{i2\pi (b+a)}{\sqrt{ab}} \left(\sum_{k=0}^{\infty} (-1)^k \frac{\left(\frac{2\sqrt{ab}}{2}\right)^{2k+1}}{k! (k+1)!} \right) \\
&= \frac{i2\pi (b+a)}{\sqrt{ab}} \mathcal{T}_1 \left(2\sqrt{ab} \right) \quad (A13)
\end{aligned}$$

Plug in $a = \frac{\zeta\kappa - t\kappa}{2}$ and $b = \frac{\zeta\kappa + t\kappa}{2}$, we have

$$\frac{1}{\pi} \int_{-\kappa}^{\kappa} \sin \zeta \sqrt{\kappa^2 - x^2} e^{-tx} dx = -\frac{2\kappa^2 \zeta}{\kappa \sqrt{\zeta^2 - t^2}} \mathcal{T}_1 \left(\kappa \sqrt{\zeta^2 - t^2} \right) \quad (A14)$$

The inverse Laplace transform of $G_2(s)$ is obtained as

$$\begin{aligned}
L^{-1} (e^{\gamma x}) &= e^{-R_b t/2L_b} L^{-1} (G_2(s)) = \frac{-R_b e^{-R_b t/2L_b}}{L_b \sqrt{x^2 L_b C_b - t^2}} x \\
&\sqrt{L_b C_b} \mathcal{T}_1 \left(\frac{R_b}{2L_b} \sqrt{x^2 L_b C_b - t^2} \right) (u(t) - u(t - \sqrt{L_b C_b} x)) \quad (A15)
\end{aligned}$$

This completes the proof of Lemma 1.

Lemma 2: $L^{-1}(A_1^i(s))$ and $L^{-1}(A_2^i(s))$ are monotonic. Besides, the sum of $L^{-1}(A_1^i(s))$ and $L^{-1}(A_2^i(s))$ is monotonic.

Proof of Lemma 2: By use of (19), the relationships between $A_1^{i-1}(s)$, $A_2^{i-1}(s)$, $A_1^i(s)$ and $A_2^i(s)$ can be formulated as

$$\begin{cases} A_1^i(s) = \frac{A_1^{i-1}(s)e^{-\gamma L_{i-1}}}{2} - \frac{A_2^{i-1}(s)e^{\gamma L_{i-1}}}{2} + \frac{\zeta}{2\gamma^2} - \frac{\zeta'}{2(\gamma')^2} \\ \quad + \frac{A_1^{i-1}(s)\gamma C_b'}{2C_b\gamma'} e^{-\gamma L_{i-1}} + \frac{A_2^{i-1}(s)\gamma C_b'}{2C_b\gamma'} e^{\gamma L_{i-1}} \\ A_2^i(s) = \frac{A_1^{i-1}(s)\gamma C_b'}{2C_b\gamma'} e^{-\gamma L_{i-1}} + \frac{A_2^{i-1}(s)\gamma C_b'}{2C_b\gamma'} e^{\gamma L_{i-1}} + \frac{A_2^{i-1}(s)e^{\gamma L_{i-1}}}{2} \\ \quad - \frac{A_1^{i-1}(s)e^{-\gamma L_{i-1}}}{2} - \frac{\zeta}{2\gamma^2} + \frac{\zeta'}{2(\gamma')^2} \end{cases} \quad (A16)$$

Suppose that pipes l_{i-1} and l_i are connected end to end. L_{i-1} is the length and ζ , γ and C_b are parameters of pipe l_{i-1} . ζ' , γ' and C_b' are parameters of pipe l_i . Taking the derivative of $L^{-1}(A_1^i(s))$ and $L^{-1}(A_2^i(s))$ with respect to t yields

$$\begin{aligned}
& [L^{-1}(A_2^i(s))]'_t = [L^{-1}(A_2^{i-1}(s))]'_t \otimes L^{-1}(e^{\gamma L_{i-1}}/2) \\
& - [L^{-1}(A_1^{i-1}(s))]'_t \otimes L^{-1}(e^{-\gamma L_{i-1}}/2) - [L^{-1}(\zeta/(2\gamma^2))]'_t \\
& + [L^{-1}(\zeta'/(2\gamma'^2))]'_t + L^{-1}\left(\frac{\gamma C_b'}{2C_b\gamma'}\right) \otimes \left\{ [L^{-1}(A_1^{i-1}(s))]'_t \right. \\
& \left. \otimes L^{-1}(e^{-\gamma L_{i-1}}) + [L^{-1}(A_2^{i-1}(s))]'_t \otimes L^{-1}(e^{\gamma L_{i-1}}) \right\} \quad (A17)
\end{aligned}$$

$$\begin{aligned}
& L^{-1}\left(\frac{\gamma C_b'}{C_b\gamma'}\right) \otimes \left\{ [L^{-1}(A_1^{i-1}(s))]'_t \otimes L^{-1}(e^{-\gamma L_{i-1}}/2) \right. \\
& \left. + [L^{-1}(A_2^{i-1}(s))]'_t \otimes L^{-1}(e^{\gamma L_{i-1}}/2) \right\} \quad (A18)
\end{aligned}$$

The operator $\otimes$ denotes the convolution operation. As t changes from 0 to $\sqrt{L_b C_b} L_{i-1}$, the range of $\sqrt{L^2_{i-1} L_b C_b - t^2}$ varies between 0 and L_{i-1}/α , and $\frac{R_b}{2L_b} \sqrt{L^2_{i-1} L_b C_b - t^2} <$

1. The Bessel function of the first kind is positive, $\mathcal{T}_1(\frac{R_b}{2L_b} \sqrt{L^2_{i-1} L_b C_b - t^2}) > 0$, and $L^{-1}(e^{\gamma L_{i-1}}) < 0$.

We also have $L^{-1}(e^{-\gamma L_{i-1}}) > 0$ and $\int_{-\infty}^{\infty} L^{-1}(e^{-\gamma L_{i-1}}/2) > -\int_{-\infty}^{\infty} L^{-1}(e^{\gamma L_{i-1}}/2)$.

The derivatives of $L^{-1}(\zeta/(2\gamma^2))$ and $L^{-1}(\zeta'/(2\gamma'^2))$ are

$$[L^{-1}(\zeta/(2\gamma^2))]' = \left[\frac{U_{sb}}{R_b} - \frac{U_{sb}}{R_b} e^{-\frac{R_b}{L_b} t} \right]' = \frac{U_{sb}}{L_b} e^{-\frac{R_b}{L_b} t} \quad (A19)$$

$$[L^{-1}(\zeta'/(2\gamma'^2))]' = \left[\frac{U'_{sb}}{R'_b} - \frac{U'_{sb}}{R'_b} e^{-\frac{R'_b}{L'_b} t} \right]' = \frac{U'_{sb}}{L'_b} e^{-\frac{R'_b}{L'_b} t} \quad (A20)$$

By [37], the inverse Laplace transform of $\gamma C_b'/(C_b\gamma')$ is given as

$$\begin{aligned}
L^{-1}\left(\frac{\gamma C_b'}{C_b\gamma'}\right) &= \frac{C'_b}{C_b} \sqrt{\frac{L_b C_b}{L'_b C'_b}} L^{-1}\left(\sqrt{1 + \frac{\frac{R_b}{L_b} - \frac{R'_b}{L'_b}}{s + \frac{R'_b C'_b}{L'_b C'_b}}}\right) \\
&= be^{-at} (I_0(at) + I_1(at)) + \delta(t). \quad (A21)
\end{aligned}$$

where $a = \frac{1}{2} \left(\frac{R_b}{L_b} - \frac{R'_b}{L'_b} \right)$ and $b = a \frac{C'_b}{C_b} \sqrt{\frac{L_b C_b}{L'_b C'_b}}$.

The following hypotheses are made:

Hypothesis 1: $[L^{-1}(A_1^{i-1}(s))]'_t > 0$, $[L^{-1}(A_2^{i-1}(s))]'_t > 0$, and $[L^{-1}(A_1^{i-1}(s))]'_t - [L^{-1}(A_2^{i-1}(s))]'_t > 0$.

Hypothesis 2: $[L^{-1}(A_1^{i-1}(s))]'_t < 0$, $[L^{-1}(A_2^{i-1}(s))]'_t < 0$, and $[L^{-1}(A_1^{i-1}(s))]'_t - [L^{-1}(A_2^{i-1}(s))]'_t < 0$.

Based on the hypothesis 1, we have $[L^{-1}(A_1^i(s))]'_t > 0$ by (A17) and $[L^{-1}(A_2^i(s))]'_t > 0$ by (A18). Add $[L^{-1}(A_1^i(s))]'_t$ and $[-L^{-1}(A_2^i(s))]'_t$, we have

$$\begin{aligned}
& [L^{-1}(A_1^i(s))]'_t - [L^{-1}(A_2^i(s))]'_t = 2 [L^{-1}(A_1^{i-1}(s))]'_t \\
& \otimes L^{-1}\left(\frac{e^{-\gamma L_{i-1}}}{2}\right) - 2 [L^{-1}(A_2^{i-1}(s))]'_t \otimes L^{-1}\left(\frac{e^{\gamma L_{i-1}}}{2}\right) \\
& + 2 [L^{-1}(\zeta/(2\gamma^2))]'_t - 2 [L^{-1}(\zeta'/(2\gamma'^2))]'_t > 0 \quad (A22)
\end{aligned}$$

Similarly, based on the hypothesis 2, we have $[L^{-1}(A_1^i(s))]'_t < 0$, $[L^{-1}(A_2^i(s))]'_t < 0$, and $[L^{-1}(A_1^i(s))]'_t - [L^{-1}(A_2^i(s))]'_t < 0$.

$A_1^1(s)$ and $A_2^1(s)$ are

$$\begin{cases} A_1^1(s) = \frac{Q_0}{2} + \frac{H_0 C_b s}{2\sqrt{L_b C_b}} \sqrt{\frac{1}{s^2 + \frac{R_b}{L_b} s}} - \frac{C_b U_{sb}}{2(R_b + s L_b) s C_b} \\ A_2^1(s) = \frac{H_0 C_b s}{2\sqrt{L_b C_b}} \sqrt{\frac{1}{s^2 + \frac{R_b}{L_b} s}} - \frac{Q_0}{2} + \frac{C_b U_{sb}}{2(R_b + s L_b) s C_b} \end{cases} \quad (A23)$$

where Q_0 and H_0 are expressed as Laplace transform of the combinations of the step functions. Since the inverse Laplace

transform of $\sqrt{1/(s^2 + \frac{R_b}{L_b}s)}$ is

$$L^{-1} \left(\sqrt{1/(s^2 + \frac{R_b}{L_b}s)} \right) = \frac{1}{2} e^{-\frac{R_b}{2L_b}t} I_0 \left(\frac{R_b}{L_b}t \right) \quad (A24)$$

It follows from (A22) to (A24) that the derivatives of $L^{-1}(A_1^1(s))$ and $L^{-1}(A_2^1(s))$ can be written

$$\begin{cases} [L^{-1}(A_1^1(s))]' = \frac{Q_1'}{2} + \frac{H_1' C_b}{2\sqrt{L_b C_b}} \otimes L^{-1} \left(\sqrt{\frac{1}{s^2 + \frac{R_b}{L_b}s}} \right) - \frac{U_{sb}}{2L_b} e^{-\frac{R_b}{L_b}t} \\ \quad \begin{cases} > 0 \text{ if } H_1' > 0, Q_1' > 0 \\ < 0 \text{ if } H_1' < 0, Q_1' < 0 \end{cases} \\ [L^{-1}(A_2^1(s))]' = \frac{H_1' C_b}{2\sqrt{L_b C_b}} \otimes L^{-1} \left(\sqrt{\frac{1}{s^2 + \frac{R_b}{L_b}s}} \right) - \frac{Q_1'}{2} + \frac{U_{sb}}{2L_b} e^{-\frac{R_b}{L_b}t} \\ \quad \begin{cases} > 0 \text{ if } H_1' > 0, Q_1' > 0 \\ < 0 \text{ if } H_1' < 0, Q_1' < 0 \end{cases} \end{cases} \quad (A25)$$

where Q_1 and H_1 are the inverse Laplace transform of Q_0 and H_0 . The difference of $L^{-1}(A_1^1(s))$ and $L^{-1}(A_2^1(s))$ is

$$\begin{aligned} [L^{-1}(A_1^1(s))]'_t - [L^{-1}(A_2^1(s))]'_t &= Q_1' \\ &- \frac{U_{sb}}{L_b} e^{-\frac{R_b}{L_b}t} \begin{cases} > 0 \text{ if } Q_1' > 0 \\ < 0 \text{ if } Q_1' < 0 \end{cases} \end{aligned} \quad (A26)$$

The hypotheses hold when $H_1' > 0, Q_1' > 0$ and $H_1' < 0, Q_1' < 0$, respectively. Lemma 2 is proved.

Theorem 1: The inverse Laplace transform of Q and H are monotonic.

Proof of Theorem 1: The derivatives of $L^{-1}(Q)$ and $L^{-1}(H)$ with respect to t are

$$\begin{aligned} [L^{-1}(Q)]'_t &= [L^{-1}(A_1^i(s) e^{-\gamma x})]'_t - [L^{-1}(A_2^i(s) e^{\gamma x})]'_t \\ &+ [L^{-1}(\zeta/\gamma^2)]'_t = [L^{-1}(A_1^i(s))]'_t \otimes L^{-1}(e^{-\gamma x}) \\ &- [L^{-1}(A_2^i(s))]'_t \otimes L^{-1}(e^{\gamma x}) + [L^{-1}(\zeta/\gamma^2)]'_t \end{aligned} \quad (A27)$$

$$\begin{aligned} [L^{-1}(H)]'_t &= \left[L^{-1} \left(\frac{A_1^i(s) \gamma}{s C_b} e^{-\gamma x} \right) \right]'_t + \left[L^{-1} \left(\frac{A_2^i(s) \gamma}{s C_b} e^{\gamma x} \right) \right]'_t \\ &= [L^{-1}(A_1^i(s))]'_t \otimes L^{-1} \left(\frac{\gamma}{s C_b} \right) \otimes L^{-1}(e^{-\gamma x}) \\ &+ [L^{-1}(A_2^i(s))]'_t \otimes L^{-1} \left(\frac{\gamma}{s C_b} \right) \otimes L^{-1}(e^{\gamma x}) = L^{-1} \left(\frac{\gamma}{s C_b} \right) \\ &\otimes \left\{ [L^{-1}(A_1^i(s))]'_t \otimes L^{-1}(e^{-\gamma x}) + [L^{-1}(A_2^i(s))]'_t \otimes L^{-1}(e^{\gamma x}) \right\} \end{aligned} \quad (A28)$$

Based on Lemma 1 and Lemma 2, the derivatives of $L^{-1}(Q)$ and $L^{-1}(H)$ with respect to t are always negative when $H_1' < 0, Q_1' < 0$, and positive when $H_1' > 0, Q_1' > 0$. The same can be obtained for the derivatives of $L^{-1}(Q)$ and $L^{-1}(H)$ with respect to x . This completes the proof of Theorem 1.

ACKNOWLEDGMENT

The authors would like to thank Prof. Kaigui Xie and Prof. Bo Hu for their valuable suggestions on improving the quality of the paper.

REFERENCES

- [1] K. Smith, "Understanding energy consumption for urban water supply in China," M.S. thesis, Dept. Environ. Sci. Eng., Tsinghua Univ., Beijing, China, 2015.
- [2] World Bank, *World Development Indicators*. Washington DC, USA: World Bank, 2016.
- [3] W. N. Lubega and A. M. Farid, "Quantitative engineering systems modeling and analysis of the energy-water nexus," *Appl. Energy*, vol. 135, pp. 142–157, Dec. 2014.
- [4] R. Menke, E. Abraham, P. Pappas, and I. Stoianov, "Extending the envelope of demand response provision through variable speed pumps," *Procedia Eng.*, vol. 186, pp. 584–591, 2017.
- [5] A. Stuhlmacher and J. L. Mathieu, "Chance-constrained water pumping to manage water and power demand uncertainty in distribution networks," in *Proc. IEEE*, vol. 99, 2020, pp. 1–16.
- [6] A. S. Zamzam, E. Dall'Anese, C. Zhao, J. A. Taylor, and N. D. Sidiropoulos, "Optimal water-power flow problem: Formulation and distributed optimal solution," *IEEE Control Netw. Syst.*, vol. 6, no. 1, pp. 37–47, Mar. 2019.
- [7] D. Fooladivanda and J. A. Taylor, "Energy-optimal pump scheduling and water flow," *IEEE Control Netw. Syst.*, vol. 5, no. 3, pp. 1016–1026, Sep. 2018.
- [8] M. K. Singh and V. Kekatos, "Optimal scheduling of water distribution systems," *IEEE Control Netw. Syst.*, vol. 7, no. 2, pp. 711–723, 2020.
- [9] R. Menke, E. Abraham, P. Pappas, and I. Stoianov, "Demonstrating demand response from water distribution system through pump scheduling," *Appl. Energy*, vol. 170, pp. 377–387, 2016.
- [10] G. Bonvin and S. Demasse, "Convex relaxation for water distribution systems," in *Proc. 28th Eur. Conf. Oper. Res.*, Poland, 2016, pp. 1–15.
- [11] Q. Li, S. Yu, A. S. Al-Sumaiti, and K. Turitsyn, "Micro water-energy nexus: Optimal demand-side management and quasi-convex hull relaxation," *IEEE Control Netw. Syst.*, vol. 6, no. 4, pp. 1313–1322, Dec. 2019.
- [12] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 1101–1110, Jan. 2019.
- [13] K. Oikonomou and M. Parvania, "Optimal coordinated operation of interdependent power and water distribution systems," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 4784–4794, Nov. 2020.
- [14] K. Oikonomou, M. Parvania, and R. Khatami, "Optimal demand response scheduling for water distribution systems," *IEEE Trans. Ind. Inform.*, vol. 14, no. 11, pp. 5112–5122, Nov. 2018.
- [15] K. Oikonomou and M. Parvania, "Optimal participation of water desalination plants in electricity demand response and regulation markets," *IEEE Syst. J.*, vol. 14, no. 3, pp. 1–11, Sep. 2020.
- [16] C. Mkireba, A. Dembélé, A. Jougleth, and T. Denoeuxb, "Robust optimization of demand response power bids for drinking water systems," *Appl. Energy*, vol. 238, pp. 1036–1047, 2019.
- [17] E. Benjaminwylie and V. Streeter, *Fluidtransients*. New York, NY, USA: McGraw-Hill International Book, 1978.
- [18] Y. Yao, C. Li, K. Xie, H.-M. Tai, B. Hu, and T. Niu, "Optimal design of water tank size for power system flexibility and water quality," *IEEE Trans. Power Syst.*, vol. 36, no. 6, pp. 5874–5888, Nov. 2021.
- [19] L. Kang and K. E. Holbert, "Applying the equivalent pi circuit to the modeling of hydraulic pressurized lines," *Math. Comput. Simul.*, vol. 79, no. 7, pp. 2064–2075, 2009.
- [20] S. Liang, "Mathematical modeling of the flow in a pipeline with a leak," *Math. Comput. Simul.*, vol. 82, no. 11, pp. 2253–2267, 2012.
- [21] R. K. Sharma, S. Mishra, and P. Barsaiyan, "Dynamic response of nonlinear electrical analogue hydroelectric turbine model," in *Proc. Annu. IEEE India Conf.*, 2015, pp. 1–6.
- [22] Q. Chen, R. Fu, and Y. Xu, "Electrical circuit analogy for heat transfer analysis and optimization in heat exchanger networks," *Appl. Energy*, vol. 139, pp. 81–92, 2015.

- [23] Q. Chen, J. Hao, and T. Zhao, "An alternative energy flow model for analysis and optimization of heat transfer systems," *Int. J. Heat Mass Transfer*, vol. 108, pp. 712–720, 2017.
- [24] S. Badakhshan, M. Ehsan, M. Shahidehpour, N. Hajibandeh, M. Shafie-Khah, and J. P. S. Catalão, "Security-constrained unit commitment with natural gas pipeline transient constraints," *IEEE Trans. Smart Grid*, vol. 11, no. 1, pp. 118–128, Jan. 2020.
- [25] J. Yang, N. Zhang, A. Botterud, and C. Kang, "On an equivalent representation of the dynamics in district heating networks for combined electricity-heat operation," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 560–570, Jan. 2020.
- [26] J. Yang, A. Botterud, N. Zhang, Y. Lu, and C. Kang, "A cost-sharing approach for decentralized electricity-heat operation with renewables," *IEEE Trans. Sustain. Energy*, vol. 11, no. 3, pp. 1838–1847, Jul. 2020.
- [27] C. Li, Y. Yao, K. Xie, B. Hu, and T. Niu, "Integrated electrical, heating, and water distribution system to accommodate wind power," *IEEE Trans. Sustain. Energy*, vol. 12, no. 2, pp. 1100–1114, Apr. 2020.
- [28] M. Alhazmi and P. Dehghanian, "Optimal integration of interconnected water and electricity networks," *IET Gener., Transmiss. Distrib.*, vol. 15, no. 14, pp. 2033–2043, 2021.
- [29] C. Wang, N. Gao, J. Wang, N. Jia, T. Bi, and K. Martin, "Robust operation of a water-energy nexus: A multi-energy perspective," *IEEE Trans. Sustain. Energy*, vol. 11, no. 4, pp. 2698–2712, Oct. 2020.
- [30] J. Sun *et al.*, "Renewable energy transmission by HVDC across the continent: System challenges and opportunities," *CSEE J. Power Energy*, vol. 3, no. 4, pp. 353–364, 2017.
- [31] M. Zhang *et al.*, "An integral identification method of characteristic parameters and optimization of parallel connection heat transfer systems based on the power flow method," *Appl. Thermal Eng.*, vol. 143, pp. 1057–1067, 2018.
- [32] S. B. Pope, *Turbulent Flows*. Cambridge, UK: Cambridge Univ. Press, 2000.
- [33] M. Shokoohi, M. Tabesh, S. Nazif, and M. Dini, "Water quality based multi-objective optimal design of water distribution systems," *Water Resour. Manage.*, vol. 31, pp. 93–108, 2017.
- [34] N. Islam, R. Sadiq, and M. Rodriguez, "Optimizing booster chlorination in water distribution networks: A water quality index approach," *Environ. Monit. Assess.*, vol. 185, no. 10, pp. 8035–8050, 2013.
- [35] J. L. Gross, J. Yellen, and M. Anderson, *Graph Theory and Its Applications*. Florida, USA: Chapman and Hall/CRC, 2018.
- [36] M. Carrión and J. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1371–1378, Aug. 2006.
- [37] F. Oberhettinger and L. Badii, *Tables of Laplace Transforms*. Berlin Heidelberg, German: Springer-Verlag, 1973.



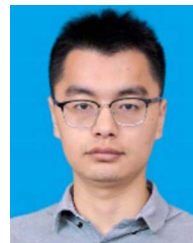
Yiming Yao received the B.S. degree in 2019 from Chongqing University, Chongqing, China, where he is currently working toward the Ph.D. degree with the School of Electrical Engineering. His research interests include energy-water nexus and renewable energy accommodation.



Chunyan Li (Member, IEEE) received the Ph.D. degree in electrical engineering from Chongqing University, Chongqing, China, in 2008. She is currently an Associated Professor with the School of Electrical Engineering, Chongqing University. Her research interests include power system analysis, renewable energy consumption, and smart grid dispatching.



Heng-Ming Tai (Senior Member, IEEE) is currently a Professor with the Department of Electrical and Computer Engineering, University of Tulsa, Tulsa, OK, USA. His research interests include power system reliability and industrial electronics.



Changzheng Shao (Member, IEEE) received the B.S. degree in electrical engineering from Shandong University, Jinan, China, and the Ph.D. degree in electrical engineering from Zhejiang University, Hangzhou, China, in 2015 and 2020, respectively. He is currently an Assistant Professor with Chongqing University, Chongqing, China. His research interests include the operation optimization and reliability evaluation of the integrated energy system.